

A Gait Foundation Model Predicts Multi-System Health Phenotypes from 3D Skeletal Motion

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Abstract

Gait is increasingly recognized as a vital sign, yet current approaches treat it as a symptom of specific pathologies rather than a systemic biomarker. We developed a gait foundation model for 3D skeletal motion from 3,328 deeply phenotyped adults, recorded via a depth camera during five motor tasks. Learned embeddings outperformed engineered features, predicting age (Pearson $r = 0.69$), BMI ($r = 0.91$), and visceral adipose tissue area ($r = 0.83$). Embeddings significantly predicted 1,887 of 3,210 phenotypic targets; after adjustment for age, BMI, VAT, and height, gait provided independent gains in all 18 body systems in males and 17 of 18 in females, and improved prediction of clinical diagnoses and medication use. Anatomical ablation revealed that legs dominated metabolic and frailty predictions while torso encoded sleep and lifestyle phenotypes. These findings establish gait as an independent multi-system biosignal, motivating translation to consumer-grade video and its integration as a scalable, passive vital sign.

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Introduction

Gait speed is increasingly recognized as a "vital sign" of overall health status ¹. Pooled analyses across nine cohorts (n = 34,485) have shown that gait speed independently predicts survival in older adults with accuracy comparable to complex multi-variable clinical models ². Beyond survival, specific gait impairments have been associated with conditions ranging from depression ³ to diabetic neuropathy ⁴ and neurodegenerative disease. ⁵

Yet despite this clinical significance, quantitative gait analysis has remained largely focused on detecting specific pathologies such as identifying the tremor of Parkinson's disease ⁶, the asymmetry of post-stroke hemiparesis ⁷, or the patterns associated with fall risk ⁸. Consequently, movement is typically treated as a symptom of dysfunction, not as a standalone biosignal for systemic health.

A key limitation impeding broader clinical utility is methodological. Most gait studies rely on engineered summary statistics, such as average cadence, stride length, and step-time variability, which reduce complex spatiotemporal sequences to a handful of scalars ⁹. While clinically validated and interpretable, this reductionist approach may discard fine-grained temporal dynamics and inter-joint coordination that encode additional systemic physiological states. ¹⁰

A systematic review of over 2,700 instrumented gait analysis studies found that more than 99% addressed only technical or condition-specific diagnostic efficacy ¹¹. To date, no prior work has systematically evaluated whether the full complexity of human gait can predict a broad panel of health phenotypes spanning multiple body systems in a large, deeply phenotyped healthy cohort.

Self-supervised learning of gait representations has recently shown promise for biomechanical estimation ¹², clinical severity scoring ¹³, and diagnostic biomarker discovery ¹⁴. However, these efforts have primarily focused on single pathologies or specific biomechanical tasks, leaving the systemic possibility of a phenome-wide gait foundation model unexplored.

Here, we adopted the Masked Autoencoder (MAE) paradigm ¹⁵, extended to skeleton sequences ¹⁶, to learn gait embeddings from 3D skeletal motion captured during five standardized motor tasks across 344 hours in 3,328 adults from the Human Phenotype Project (HPP) ¹⁷. To our knowledge, this represents the largest 3D skeletal motion dataset by subject count. Critically, movement activity - which we refer to as 'gait' throughout this study - was recorded with a single depth camera within a commercially available clinical movement laboratory, suggesting that meaningful health signals could be extracted from accessible recording setups.

We evaluated these embeddings against a comprehensive set of phenotypic targets spanning 18 body systems, including cardiovascular function, serum metabolomics, sleep, bone density, liver health, and mental health indices. By benchmarking against demographic and anthropometric baselines, specifically: age, body mass index (BMI), and visceral adipose tissue (VAT) ¹⁸, we demonstrate that gait embeddings provide independent predictive information across metabolic, cardiovascular, hematopoietic, and psychiatric phenotypes. Furthermore, anatomical ablation analysis reveals biologically interpretable patterns linking specific body regions to distinct health outcomes, establishing gait as a multi-system biomarker of human health.

Results

Deep phenotyping of the study cohort

We analyzed data from the Human Phenotype Project (HPP), a longitudinal study linking deep physiological profiling with multi-omics data. The study population consisted of 3,328 adults (aged 20–79 years) residing in Israel, predominantly of Ashkenazi (European) Jewish descent but also including participants of North African, Near Eastern, and other ancestries. Participants were generally healthy at enrollment, with exclusion criteria applied for severe active disease (**Methods**).

This cohort includes extensive clinical, behavioral, physiological and multi-omics profiling categorized here into 18 body systems (Figure 1a). We focused on analyzing multi-activity gait recording data from 3,328 participants to investigate their associations with concurrent phenotypic measurements. Throughout this work, we use 'gait' broadly to encompass the full movement assessment protocol, which includes both locomotor tasks and complementary postural and functional motor assessments. The final analytical cohort included 1,607 males and 1,721 females with a mean age of 52.5 ± 10.1 years and mean BMI of $25.7 \pm 3.8 \text{ kg m}^{-2}$ (**Extended Table 1**).

Reference values and limitations of traditional gait descriptors

Gait activities consisted of five standardized motor tasks, recorded using the Movement Lab (Newton Tech), a depth camera-based movement analysis system, designed to probe complementary aspects of locomotor and postural control (see **Methods, Data Acquisition**).

Data and model exploration

Analysis of these conventional gait descriptors revealed pronounced sexual dimorphism. During fixed-speed treadmill walking, females exhibited higher cadence (100.8 vs. 94.5 steps min^{-1}) and shorter stride length (0.98 vs. 1.05 m) compared with males. Functional task performance also differed by sex, with females demonstrating shorter chair-rise times during sit-to-stand (657 vs. 711 ms), whereas males exhibited larger anteroposterior sway amplitudes during the Romberg test (234 vs. 216 mm).

Despite their ability to capture gross kinematic differences, these engineered summary metrics showed limited utility for biological inference. Models trained exclusively on traditional gait descriptors failed to explain substantial variance in basic demographics such as chronological age and body mass index (BMI) (**Figure 2**).

Extensive benchmarking across the full phenotypic panel revealed that these descriptors yielded modest predictive gains above a basic demographic baseline (age, BMI) in a limited number of body systems (**Extended Figure 1**). These results indicate that while conventional gait descriptors capture some health-relevant variation, their sensitivity remains limited compared with the learned representations described below.

Self-supervised learning of gait dynamics

To capture the rich temporal and spatial structure of human movement, we employed a representation learning approach based on a dual-stream spatiotemporal transformer architecture trained directly on raw 3D skeletal motion sequences. Model training was performed using a denoising masked autoencoding objective, in which the network reconstructed partially observed motion sequences under combined spatial (joint-level) and temporal masking (**Methods - Deep learning section**). The trained model achieved a low reconstruction error (mean per joint position error of 0.008), corresponding to an average joint position error of approximately 8 mm, indicating accurate recovery of masked gait dynamics. (**Figure 1c**)

From the trained network, we extracted compact gait embeddings summarizing both spatial configuration and temporal coordination patterns across joints. These embeddings enabled robust prediction of chronological age across individual motor tasks. Among single-task models, treadmill walking at 3 km h⁻¹ achieved the strongest performance (Pearson $r = 0.635 \pm 0.003$), while other tasks such as the sit-to-stand task remained predictive (Pearson $r = 0.536 \pm 0.002$; **Figure 2c,d**). To integrate complementary information across tasks, we constructed a Gait Fusion model by averaging predictions derived from each activity-specific embedding. This ensemble substantially improved age prediction performance (male: Pearson $r = 0.691 \pm 0.002$, female: Pearson $r = 0.689 \pm 0.002$), significantly outperforming models based on conventional gait descriptors combined with height (male: Pearson $r = 0.455 \pm 0.007$, female: Pearson $r = 0.464 \pm 0.006$).

The same representations captured body composition with high fidelity. Regression on BMI using the fused model yielded strong correlations in both sexes (male: Pearson $r = 0.896 \pm 0.0$, female: Pearson $r = 0.915 \pm 0.002$, **Figure 2e,f**). To assess relevance to metabolically meaningful adiposity, we further regressed gait embeddings against visceral adipose tissue (VAT) area, achieving male Pearson $r = 0.824 \pm 0.001$, female Pearson $r = 0.817 \pm 0.002$, compared with Pearson $r = 0.488 \pm 0.010$ and 0.423 ± 0.007 respectively using conventional gait features and height (**Figure 2g,h**). Gait embeddings also discriminated sex with near-perfect accuracy (ensemble AUC = 0.99 ± 0.002 ; >0.95 for all individual tasks), motivating the sex-stratified analyses reported below. Additionally, Gait embeddings captured individual-specific signatures that were temporally stable across repeat visits (**Supplementary Figure 1**).

We compared the embeddings against two baselines. Static inter-keypoint distances from a single frame, with no temporal information, predicted BMI at $r = 0.38 / 0.37$ (male / female), VAT at $0.27 / 0.33$, and age at $0.07 / 0.05$. Pooled normalized keypoints, retaining time but no learned representation, predicted through linear and non-linear models, predicted BMI at $r = 0.86 / 0.88$, VAT at $0.77 / 0.77$, age at $0.58 / 0.54$. (**Extended Figure 2**). Self-supervised pretraining adds little where body geometry already suffices and most where it does not.

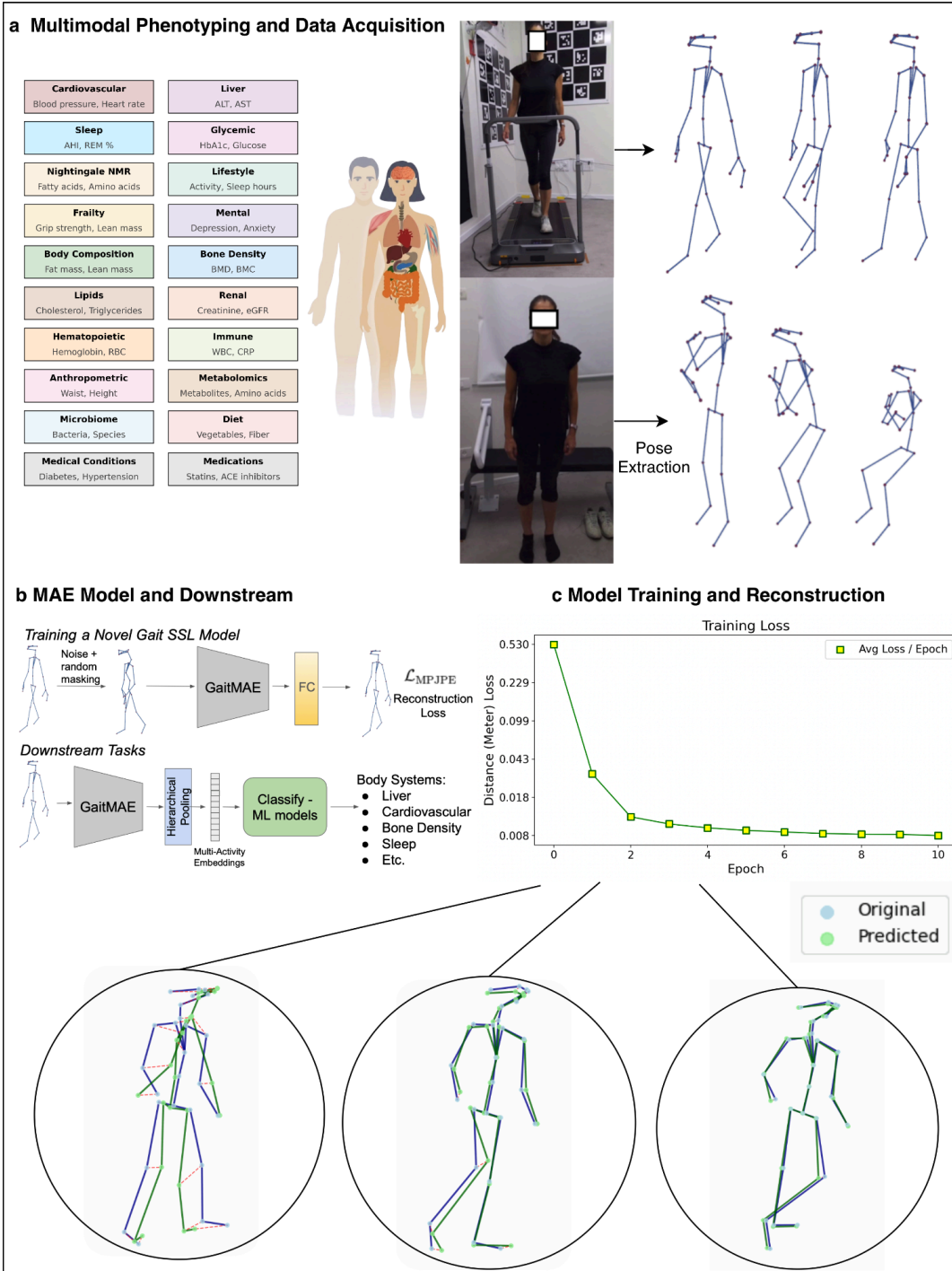


Figure 1: Overview of the Gait Foundation Model, data acquisition, and training pipeline. **a**, Multimodal phenotyping and gait acquisition. Left: The study cohort (Human Phenotype Project) underwent deep profiling across 18 distinct physiological systems (color-coded categories), ranging from cardiovascular and metabolic markers to mental health and lifestyle indices. Right: Movement data were acquired using a single Azure Kinect depth camera during five standardized motor tasks (for example, treadmill walking and sit-to-stand). Raw depth data was processed to extract 3D skeletal pose sequences. **b**, Self-supervised architecture and downstream evaluation. Top: The Gait foundation model is pre-trained using a denoising masked autoencoder objective. Input skeleton sequences are corrupted with Gaussian noise and random spatio-temporal masking; the model is trained to

reconstruct the original motion, minimizing the mean per-joint position error down to 0.008 (kin to 8mm error). Bottom: For downstream tasks, representations from the frozen encoder undergo hierarchical pooling to generate subject-level multi-activity embeddings, which are then used to train standard machine learning models for phenotype prediction. **c**, Model training and reconstruction performance. Top: Training loss curve (mean distance error in meters) showing convergence to a reconstruction error of approximately 8 mm. Bottom: Qualitative visualization of reconstruction fidelity, displaying the overlap between ground truth (Original; blue) and model-reconstructed (Predicted; green) skeletons. ACE, angiotensin-converting enzyme; ALT, alanine transaminase; AST, aspartate transaminase; BMI, body mass index; BMD, bone mineral density; CRP, C-reactive protein; eGFR, estimated glomerular filtration rate; HbA1c, glycated hemoglobin; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; RBC, red blood cells; REM, rapid eye movement; WBC, white blood cells.

Gait embeddings alone predict phenotypes across 18 body systems

To identify key associations within other body systems, we explored the prediction power of our Gait Fusion model on features from all other body systems. This analysis simulates a passive screening setting in which only a gait recording is available, with no accompanying clinical or demographic information. We extracted gait feature embeddings for each subject in our study cohort and trained with linear and non-linear models for each phenotype. In total, we found 1,887 significant associations out of 3,210 phenotypes tested after False Discovery Rate (FDR) correction.

Predictive performance was strongest for structural and frailty metrics: body composition, bone density, and frailty metrics (Lean mass and hand grip strength) came out as most significant. In addition, engineered gait features were also strongly predicted, serving as a technical validation. Sleep and Cardiovascular measures were predicted to a substantial extent compared to other body systems.

Notably, organ-specific function was also captured, with liver function features such as liver sound speed and liver elasticity were relatively strongly predicted. Our Lifestyle features come from questionnaires and as such remain elusive to high predictions but gait most strongly predicted walking habits and exercise out of all lifestyle questions (**Figure 3**). In addition, Gait exhibited connections to the molecular level demonstrating significant prediction of cholesterol, hemoglobin and various blood tests connections.

Stratification by sex confirmed that these predictive trends were broadly concordant across sexes, though notable divergences emerged (**Extended Figure 3**). Frailty, body composition, and dietary features were more strongly predicted in males, whereas cardiovascular and liver phenotypes showed higher predictive performance in females. Additionally, we quantified how much gait data these predictions require by scaling recording length from 2 to 180 seconds. Body-composition targets reached most of their performance within the shortest windows, while dynamic targets continued to improve with longer recordings (**Extended Figure 4**). Intermittent sampling performed as well as, and in some cases better than, continuous recording of equal total length, indicating that robust prediction does not require a single uninterrupted capture.

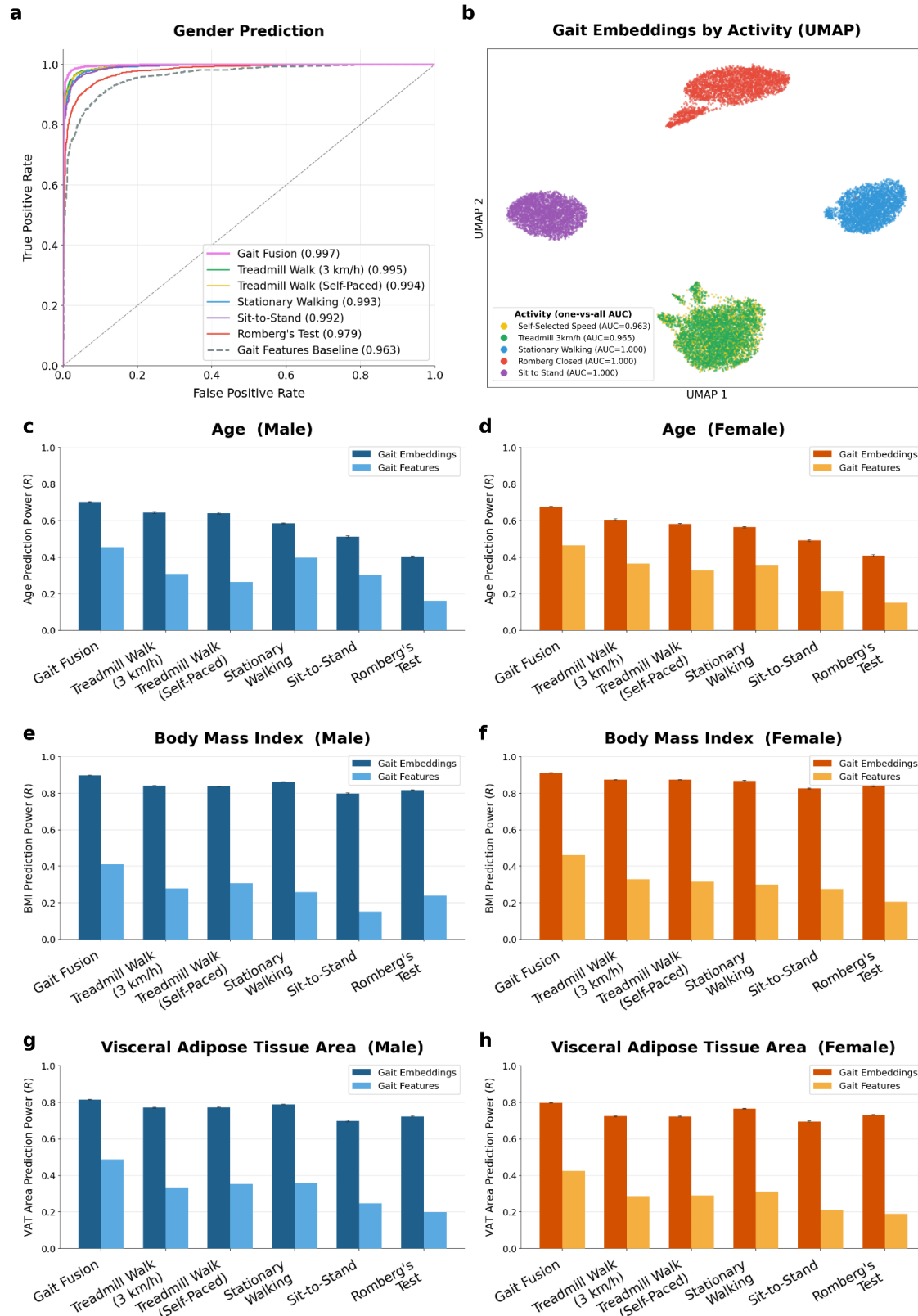


Figure 2 | Gait embeddings encode biological characteristics with high fidelity.

a, Sex classification performance evaluated by ROC curves across individual motor tasks and the Gait Fusion ensemble. All embedding-based models exceed $AUC = 0.91$, with Gait Fusion achieving $AUC = 0.997$. The engineered gait features baseline (dashed) achieves $AUC = 0.963$. b, UMAP projection of gait embeddings colored by motor task, showing distinct activity-specific clusters. One-vs-all AUC values confirm task separability ($AUC \geq 0.962$). c,d, Chronological age prediction (Pearson r) for male (c) and female (d) participants, comparing learned gait embeddings against engineered gait features (cadence, stride length, sway metrics, and height) across individual tasks and the Gait Fusion ensemble. e,f, BMI prediction for male (e) and female (f) participants. g,h, Visceral adipose tissue (VAT) area prediction for male (g) and female (h) participants. Across all targets and both sexes, learned embeddings (dark bars) substantially outperform engineered features (light bars), with the Gait Fusion ensemble consistently achieving the highest correlation. All predictions were evaluated using 5-fold nested cross-validation repeated across 15 random seeds ($n = 75$ train-test splits), with sex stratification applied throughout (Methods).

Adjusting for Covariates Across Body System-Level Associations

While direct prediction of phenotypes supports the utility of gait for accessible screening, these results raise a critical question: do gait embeddings capture distinct biological signals, or do they merely proxy well-established confounders? Our model predicts age, BMI, and sex with high accuracy (**Figure 2a–b**), and these variables are known to drive a wide range of health phenotypes^{19,20}. Moreover, VAT, a clinically meaningful marker of metabolic risk²¹, is also predicted with high fidelity. It is therefore plausible that the observed phenotypic associations are mediated entirely through these factors.

To address this, we employed a predictive gain approach: for each phenotype, we compared the predictive performance of demographic and anthropometric confounders alone (age, BMI, height²², VAT) against confounders combined with gait embeddings extracted from the foundation model. Both models were trained with linear and non-linear models and were evaluated within a nested cross-validation framework. All analyses were further stratified by sex to prevent sex-linked gait variation from being conflated with phenotype-relevant signals. Performance was evaluated using Pearson r , and the improvement (Δr) attributable to gait is reported across body systems (**Methods - predictive models**).

Gait embeddings consistently yielded incremental predictive power across body systems, indicating that movement encodes health information orthogonal to body habitus and age (Figure 4). In males, gait embeddings significantly improved prediction in all 18 body systems; in females, 17 of 18 systems showed significant gains, with renal function as the sole exception. The largest median improvements were observed in hematopoietic markers, mental health phenotypes, and frailty indices. Metabolomics and liver markers also showed substantial gains, while bone density, despite a moderate median improvement, exhibited the widest spread, with several individual phenotypes reaching Δr values above 0.15. Cardiovascular and body composition phenotypes showed more modest but consistent gains. Even systems with smaller median deltas, such as sleep and metabolomics, contained individual phenotypes with notable improvements, suggesting gait captures information relevant to diverse physiological processes. These gains were stable across age and BMI strata in both sexes, with the body-system ranking and magnitude of Δr broadly preserved within subgroups (Supplementary Figure 2). We benchmarked the gait gain on liver elasticity against liver blood tests. The gait increment over an age and BMI baseline ($\Delta r = +0.07$ in males, $\Delta r = +0.03$ in females) matched or slightly exceeded the gain from adding the full liver

blood tests panel (alkaline phosphatase, ALT, AST, total protein, total bilirubin, platelets, albumin; $\Delta r = +0.06$ in males, $\Delta r = +0.01$ in females) to the same baseline (Extended Figure 5). Some paired follow-up data were available for a limited subset ($n=554$) which we report in Supplementary Figure 3. Together, these results position gait not as a proxy for age or body habitus, but as an independent biosignal with broad clinical relevance (**Figure 4**).

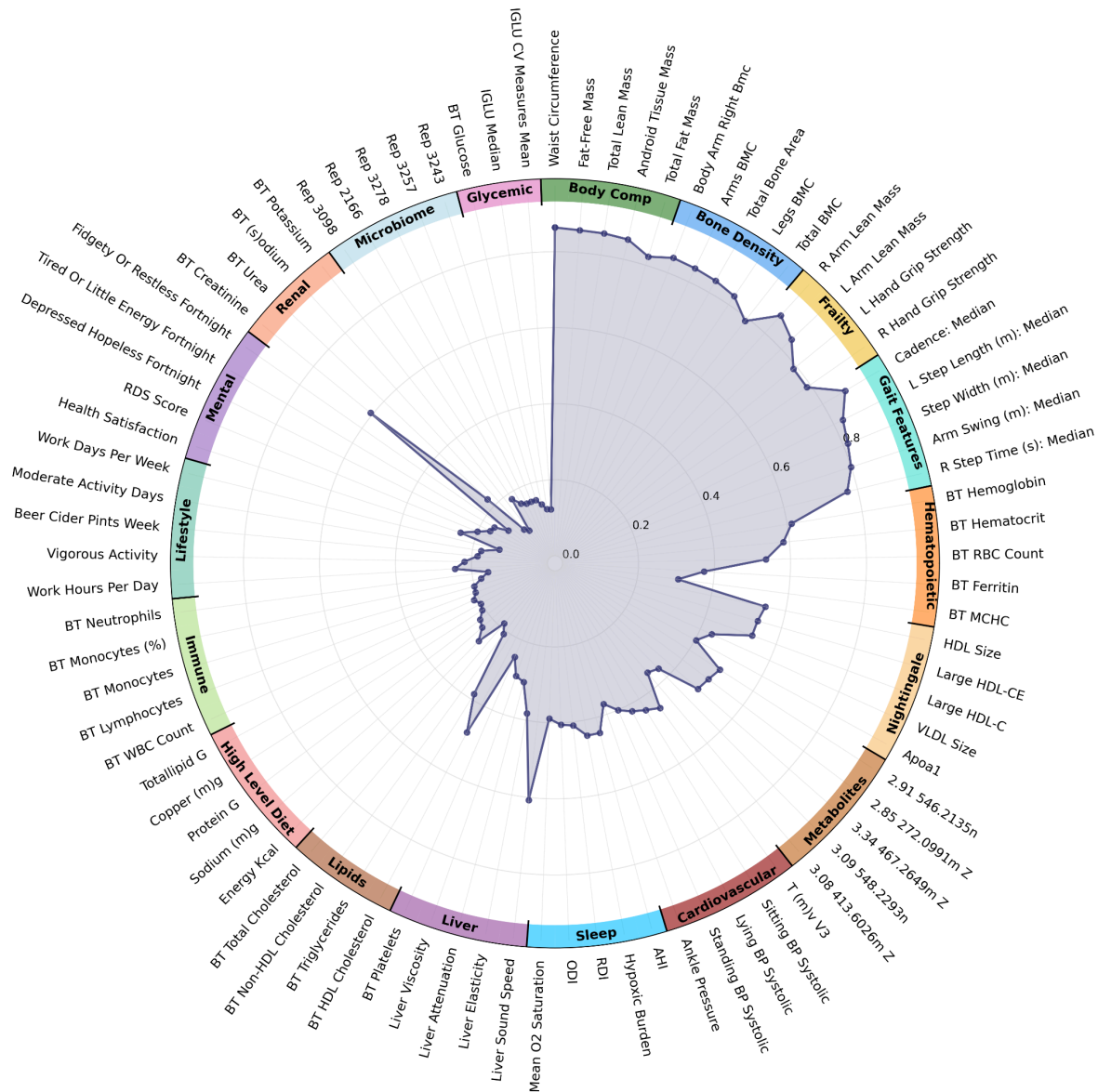


Figure 3 | Gait embeddings independently predict phenotypes across diverse body systems. Radar plot showing the top 5 features most strongly predicted by the Gait Fusion model for each body system (FDR-corrected $P < 0.05$). This analysis simulates a passive screening scenario in which only a gait recording is available, with no demographic, anthropometric, or clinical covariates included as model inputs, equivalent to predicting a person's health profile solely from filming them as they walk. Each slice represents a single phenotype, grouped by body system (labeled in the inner ring). The radial axis spans Pearson r correlation of predictions from 0 to 1. Features within each body system are ordered clockwise.

Predictions were generated using linear ridge regression on gait embeddings from the frozen Gait Foundation model encoder, evaluated via 5-fold nested cross-validation repeated across 15 random seeds ($n = 75$ train–test splits). AHI, apnea–hypopnea index; ALT, alanine transaminase; AST, aspartate transaminase; BMC, bone mineral content; BT, blood test; HDL, high-density lipoprotein; HDL-C, high-density lipoprotein cholesterol; HDL-CE, high-density lipoprotein cholesterol ester; IGLU, indices of glycemic variability; MCHC, mean corpuscular hemoglobin concentration; ODI, oxygen desaturation index; RDI, respiratory disturbance index; RBC, red blood cells; RDS, recent depressive symptoms; VLDL, very-low-density lipoprotein; WBC, white blood cells.

Gait signatures correlate with clinical diagnoses and medication use

To evaluate the clinical relevance of gait beyond continuous metabolic markers, we examined the associations between gait embeddings and a broad panel of medical diagnoses and medication usage. Analyses were stratified by sex and performed using logistic regression models rigorously adjusted for age, BMI, and VAT (**Methods**).

Gait embeddings contributed significant independent predictive information across a broad spectrum of conditions, substantially exceeding the baseline in nearly all cases (**Figure 5**). In male participants, the strongest predictive gains appeared for depression, glaucoma, insomnia, back pain, and peptic ulcer disease. These associations were corroborated by predictable usage of related medication categories, including proton pump inhibitors (PPIs), antihistamines, and antidepressants. In female participants, the ensemble model showed particularly strong performance for irritable bowel syndrome (IBS), peptic ulcer disease, psoriasis, headache, depression, and diabetes. Predictable medication usage again mirrored these conditions, with strong associations found for adrenergic inhalants, iron supplements, and hormone antagonists.

Sex-specific patterns were evident across both conditions and medications. Male participants showed unique associations with heart valve disease, vitiligo, and sinusitis, whereas female participants exhibited distinct associations with gynecological conditions, retinal detachment, and insomnia. Medication patterns mirrored this divergence, antidepressant use was predictable from gait in both sexes, while iron supplement use was specific to female participants²³

To assess whether these associations translate into screening yield, we evaluated sensitivity at a fixed top-10% referral threshold, comparing an age, BMI, and height baseline against the same covariates plus gait embeddings (Extended Figure 6). Adding gait raised sensitivity for the majority of significant conditions in both sexes.

participants. Bottom panels show the median Pearson correlation r between predicted and actual values for models combining gait embeddings with age, BMI, VAT and height. Top panels show the predictive gain (Δr) attributable to gait embeddings relative to the baseline model trained on age, BMI, VAT and height alone. Predictive power was evaluated using 5-fold nested cross-validation repeated across 15 random seeds ($n = 75$ train-test splits). The number of features with significantly improved predictions when adding gait embeddings (two-sided t-test, FDR-corrected $P < 0.1$), corresponding to the number of dots (n) included in each distribution, is shown for each body system in brackets on the x axis as (n /total number of features in the body system). Body systems are ordered from left to right by descending median Δr (top panels). b,d, Horizontal bar and dot plots ($n = 15$ seeds; bar, mean; bar, mean; whiskers, s.d) comparing the performance of the baseline model (age, BMI, VAT and height; light bars) versus the full model (gait embeddings combined with age, BMI, VAT and height; dark bars) for selected features per body system, for male (b) and female (d) participants. Features were selected to highlight clinically and physiologically notable associations across body systems, prioritizing phenotypes with substantial predictive gain (Δr) and moderate baseline performance (Pearson $r < 0.6$). Features are grouped and color-coded by the body system. Individual cross-validation Pearson r values are plotted as dots. All analyses were sex-stratified throughout. AHI, apnea-hypopnea index; ALT, alanine transaminase; BMC, bone mineral content; ECG, electrocardiogram; HCT, hematocrit; HDL, high-density lipoprotein; MCHC, mean corpuscular hemoglobin concentration; MCV, mean corpuscular volume; RDS, recent depressive symptoms; VLDL, very-low-density lipoprotein; WBC, white blood cells; S1P, sphingosine-1-phosphate. PUFA, poly unsaturated fatty acids;

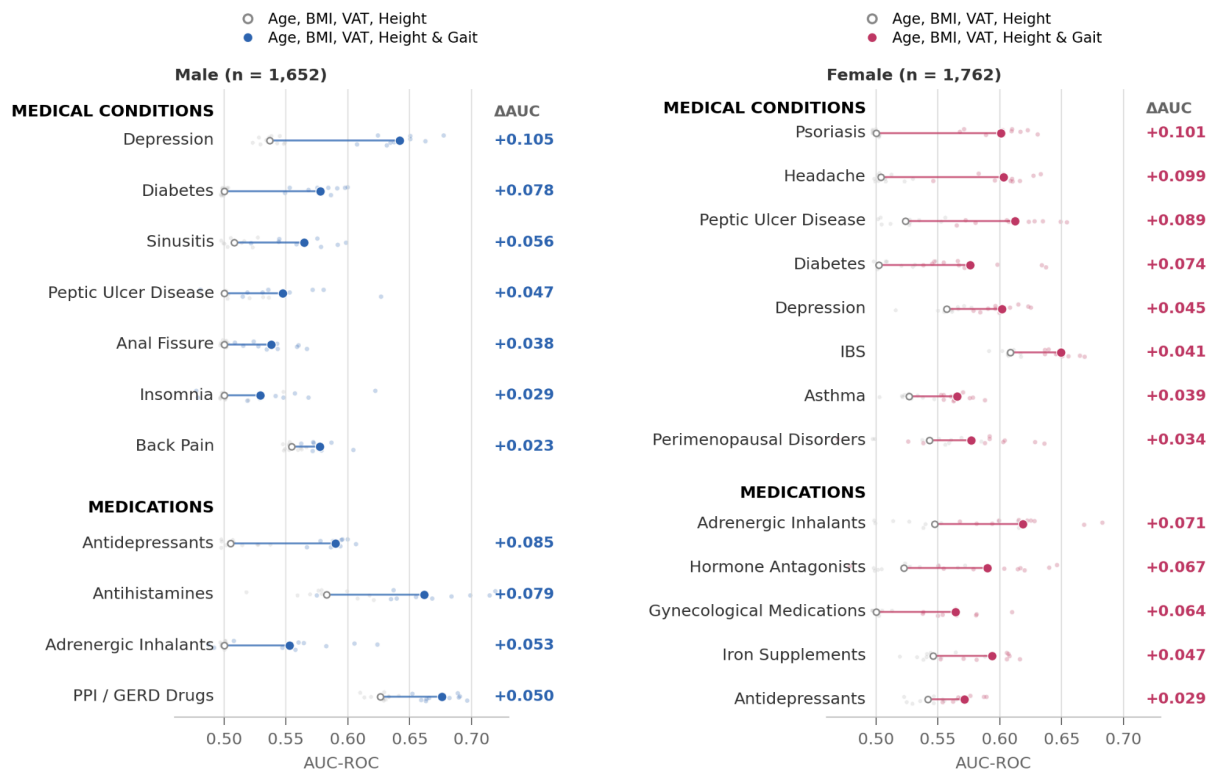


Figure 5 | Gait embeddings improve prediction of clinical diagnoses and medication use beyond demographic and anthropometric baselines.

a,b Dumbbell plots comparing AUC-ROC for models trained on covariates alone (age, BMI, VAT, height, open circles) versus models combining covariates with gait embeddings (filled circles) for male (**a**) and female (**b**) participants. Top sections show medical conditions; bottom sections show medication categories. The predictive gain (Δ AUC) attributable to gait is annotated for each target. Conditions and medications were selected as the top entries by Δ AUC within each sex. All models used logistic regression with ridge regularization, evaluated via 5-fold nested cross-validation repeated across 15 random seeds ($n = 75$ train-test splits). Significance was assessed by two-sided t -test comparing full versus baseline model AUC distributions (FDR-corrected $P < 0.1$). Analyses were sex-stratified throughout. IBS, irritable bowel syndrome; PPI, proton pump inhibitor; GERD, gastroesophageal reflux disease.

Interpretability Analysis

To investigate which anatomical regions drive the model's health predictions, we conducted body-part ablation studies. We defined four joint groups - head, arms, torso, and legs; and computed an importance score for each group per health outcome by combining two complementary ablation strategies: measuring the performance drop when masking a group from the full skeleton, and measuring each group's isolated predictive contribution relative to other groups. This analysis was restricted to treadmill walking activities, as comparing anatomical importance across task types would conflate task-specific biomechanical demands with health-relevant signals. Scores were normalized to [0, 1] and averaged across the top features per body system (see **Methods**). Full results are presented in Figure 6; here we highlight the most notable patterns.

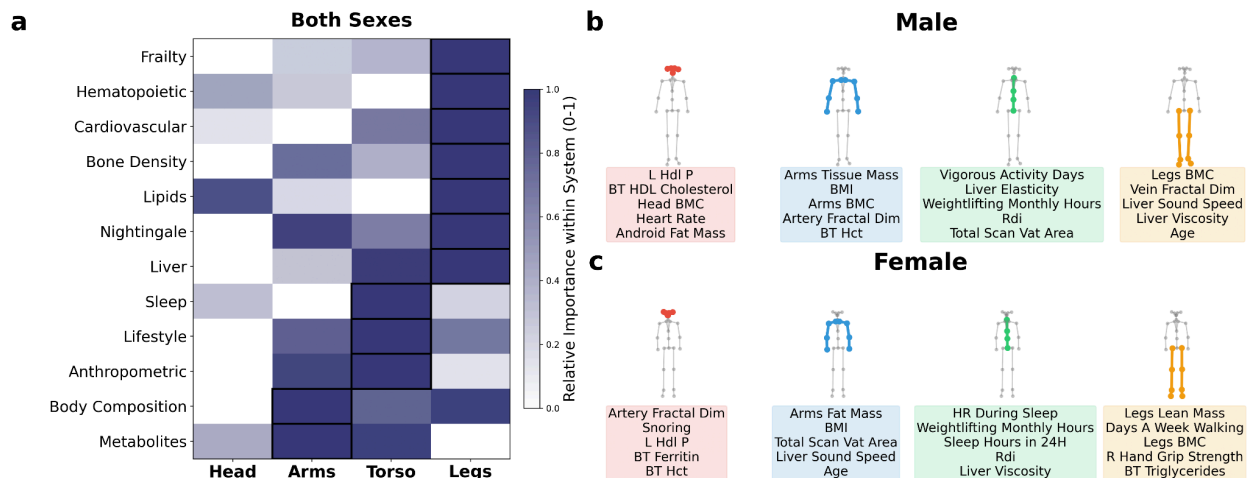


Figure 6 | Interpretability of gait embedding predictions reveals anatomically specific health associations.

a, Heatmap showing the mean importance score of the top 10 significantly predicted labels per body system across four anatomical joint groups (head, arms, torso, legs). Importance scores were computed by combining two complementary ablation strategies: the performance drop when masking a group from the full skeleton and each group's isolated predictive contribution relative to other groups. Scores were normalized to [0, 1] and averaged across the top 10 features per body system. Sex was included as a covariate to capture importance patterns across both sexes. Legs emerged as the dominant predictive region across most body systems, while sleep and lifestyle phenotypes were most strongly associated with torso dynamics. **b,c**, Sex-stratified anatomical attribution for five representative health outcomes per joint group in male (**b**) and female (**c**) participants. Each skeleton highlights the joint group (color-coded) most important for the listed phenotypes. BMC, bone mineral content; BMD, bone mineral density; BT, blood test; ECG, electrocardiogram; Hct, hematocrit; MCHC, mean corpuscular hemoglobin concentration; R, right; VAT, visceral adipose tissue area.

The legs emerged as the most important joint group for 9 of 12 body systems (**Figure 6a**) and were the top-ranked region for more individual phenotypes than any other group in both sexes (51 of 155 labels in males; 59 of 155 in females). This is consistent with the central role of lower-limb kinematics in conventional gait analysis²⁴⁻¹¹. Beyond standard locomotor metrics, leg dynamics were important for predicting general frailty markers such as grip strength, hematopoietic parameters including hemoglobin and hematocrit, and serum lipids such as triglycerides, in both sexes. Leg importance also extended to behavioral phenotypes, with self-reported weekly walking duration among the lifestyle features most strongly attributed to this region.

Leg dynamics alone carried much of the predictive signal but not all of it. A model pretrained only on the lower-limb joints recovered most of full-body performance for the body-composition anchors: BMI ($r=0.73$ vs 0.90 , males; 0.78 vs 0.91 , females) and VAT ($r=0.66$ vs 0.82 , males; 0.69 vs 0.82 , females). The gap widened for chronological age ($r=0.51$ vs 0.69 , males; 0.48 vs 0.69 , females). Across body systems, lower-limb input reached significance for 91.0% of full-body-significant phenotypes in males and 84.1% in females, but retained less of the effect magnitude (per-system median Δr retention 70% in males, 48% in females). The legs thus detect most of what the full skeleton detects, while the full body carries more of the signal strength, particularly in females. (Supplementary Figure 4)

As expected, anthropometric and body composition targets showed distributed importance across all four anatomical regions (**Figure 6a**), reflecting the whole-body nature of these measurements. An unexpected finding was that the torso joint group carried the highest importance for both sleep and lifestyle body systems (**Figure 6a**). The torso may therefore serve as a convergence point in which both sleep quality and exercise habits manifest as measurable changes in trunk dynamics during gait.

Several phenotypes showed distinct anatomical attributions between sexes (**Figure 6b,c**). Most notably, chronological age prediction was driven predominantly by leg dynamics in males but by arm and torso dynamics in females. Bone mineral density (BMD) also exhibited sex-specific patterns: in females, both total and regional BMD were most strongly associated with the head joint group, whereas in males BMD importance was distributed more evenly across regions. These sex-divergent attribution patterns suggest that the biomechanical signatures of aging and skeletal health are not uniform across sexes, an observation we explore further in the Discussion.

Taken together, the anatomical attribution analysis reveals that gait embeddings do not rely on a single movement signature but instead encode region-specific biomechanical patterns that map onto distinct physiological systems. The observation that different body regions predict different health domains suggests that targeted motor assessments, such as trunk-focused tasks for sleep and lifestyle screening or lower-limb protocols for metabolic and frailty markers, could be designed to improve the sensitivity of gait-based health monitoring for specific clinical applications.

Discussion

For this study, we analyzed 17,589 gait sequences from 3,328 deeply phenotyped adults in the Human Phenotype Project, applying self-supervised deep learning to extract representations from 3D skeletal motion captured with a single depth camera ¹⁴. Gait biomarker predictions span 18 body systems and survive adjustment for age, BMI, visceral adipose tissue, and height. As such, we propose a paradigm shift: from analyzing gait as a symptom revealer through deviations in engineered summary statistics ²⁵, to treating gait as an independent multi-system biomarker revealed through deep neural networks.

This shift is grounded in a fundamental representational difference. While conventional gait descriptors have proven clinically valuable ²⁶, and while engineered features in our own cohort successfully predicted several biological measures, these descriptors reduce complex spatiotemporal sequences to a handful of scalars, potentially discarding inter-joint coordination and temporal dynamics that encode systemic physiological state. In our cohort, models trained on engineered features yielded significant predictive gains in a some of body systems but across the majority of the body systems failed to improve prediction beyond a basic demographic baseline (Extended Figure 2), whereas learned embeddings yielded significant gains in all 18 systems in males and 17 of 18 in females (Figure 4). The performance gap was most striking for targets with no obvious kinematic correlate, including liver, hematopoietic markers, lifestyle and mental health phenotypes, suggesting that the information captured by deep representations is qualitatively different from what summary statistics can access.

A key advantage of our approach is its clinical accessibility. All gait recordings were acquired using a single front-facing depth camera via a commercially available movement analysis system (Movement Lab, Newton Tech), requiring no wearable sensors, force plates, or specialized gait laboratory infrastructure. This stands in contrast to most instrumented gait analysis studies, which rely on multi-camera motion capture systems or pressure-sensitive walkways that limit deployment to specialized centers ¹¹. Moreover, architectures closely related to ours have demonstrated successful reconstruction of 3D pose from monocular 2D video ²⁷, raising the possibility that comparable health-relevant embeddings could be extracted from standard smartphone or clinic-mounted cameras ^{28,29}. If validated, this would enable gait-based health screening at population scale with effectively zero marginal hardware cost.

A central finding of this work is that gait embeddings provide predictive information independent of age, body mass index (BMI), visceral adipose tissue (VAT), and height. Our model predicts VAT with high fidelity ($r = 0.83$), yet gait continues to improve phenotype prediction even after VAT is included as a covariate. We note that the highest raw correlations, for body-composition targets such as BMI and VAT, reflect anthropometry more than learned gait dynamics. Most of this signal is recoverable without the learned representation: a pooled-keypoint baseline that retains time but no learned representation predicts BMI and VAT nearly as well as the embeddings, and a substantial fraction is recoverable from body

proportions in a single static frame alone. Self-supervised pretraining adds little for these targets. Its contribution is largest for phenotypes with no trivial anthropometric correlation, such as age, heart rate, and liver elasticity, where neither baseline approaches the learned embeddings. For these markers, gait may carry functional information about systemic state that body size alone does not capture, accessible only through learned representations rather than engineered descriptors.

Among the most notable examples is the prediction of liver elasticity and sound speed beyond the VAT-adjusted baseline. The fact that gait adds predictive value for these markers is quite striking, as VAT itself is one of the strongest non-invasive correlates of Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD) severity³⁰. A plausible explanation lies in the bidirectional relationship between the liver and skeletal muscles.³¹ Consistent with this, reduced gait speed has been associated with advanced liver fibrosis and MASLD^{32,33}. Our findings suggest that liver health leaves a detectable imprint on gait beyond what baseline covariates capture. This warrants further investigation in both clinical and healthy populations. We also note the prediction of depression beyond demographic baselines in both sexes is consistent with well-established bidirectional associations between depressive symptoms and altered gait patterns³. The fact that antidepressant use was also captured by gait is noteworthy: some major classes of antidepressants, including Selective Serotonin Reuptake Inhibitors (SSRIs), Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs), and tricyclic antidepressants, have been associated with movement disorders such as parkinsonism, akathisia, dystonia, and tremor.^{34,35} Our study reflects these associations and reveals further links to mental health, including self-reported happiness level, suggesting gait may serve as an indicator of mood state.

The clinical value of gait here rests not on raw correlation magnitude but on the value it adds to a population wide diagnostic. The translation becomes concrete when these gains are expressed as screening yield. At a fixed top-10% referral threshold, adding gait to the model raised the fraction of true cases captured for most conditions. For example, adding gait to a model with age, BMI and height increased the fraction of true cases captured by 81% for depression and 83% for diabetes in males, and by 168% for peptic ulcer disease in females above that same model (Extended Data Figure 6). Several of these conditions were corroborated by their associated medication use, with conditions including peptic ulcer disease predicted alongside proton pump inhibitor use and asthma predicted alongside adrenergic inhalants (Figure 5). Benchmarking of these seemingly small gains revealed them to be clinically influential. One example is liver elasticity, the primary non-invasive measure of liver fibrosis³⁶. Our data reveals that gait carries information equivalent to the gain of standard liver blood tests (Extended Figure 5). Concordantly, other multi-system work reports gains of similar magnitude over demographic baselines and establish that this translates to clinical utility³⁷. This is the frame in which the effect sizes should be read. In a deeply phenotyped cohort, age and BMI already predict much of the explainable variance, and discrimination metrics are known to be insensitive to new predictors once a baseline already predicts well³⁸. Small increments over strong baselines often translate to meaningful clinical value³⁹. This positions gait as an accessible and non-invasive biomarker capable of flagging individuals that warrant a follow-up test.

Our interpretability analysis revealed that both sleep and lifestyle phenotypes were most strongly driven by trunk dynamics rather than limb kinematics, which is biologically plausible: sleep deprivation has been shown to impair postural control and increase trunk sway^{40,41}, and habitual physical activity improves core neuromuscular control and trunk stability^{42,43}. Beyond region-specific patterns, the anatomical

attribution also revealed notable sex differences: age prediction was driven by leg dynamics in males but by arm and torso kinematics in females, consistent with evidence that males experience greater rates of lower-limb strength decline with age ⁴⁴. Consequently, lower-limb kinematics were most influential out of the four anatomical groups we interpreted. A model trained on lower-limb joints - simulating leg wearables data - recovered many of the health-relevant associations observed with full-body motion (Supplementary Figure 4). However, the full-body model recovered stronger predictive associations across a broader range of physiological systems. Additionally, our analysis (Extended Figure 2) and additional literature ⁴⁵ motivate longitudinal acquisition rather than reliance on a single short recording. Therefore we suggest balancing the collection of camera-based capture with wearables data, which can also help with camera limitation and imprecision such as occlusion from real world collection or physiotherapist assisted walking. We believe an important direction will be to learn 3D-skeleton-like representations directly from wearable signals, bringing aspects of whole-body motion modeling to continuously deployable sensors. A practical system may therefore combine periodic full-body recordings, which provide a broad map of systemic state, with wearable monitoring that supplies longitudinal coverage and remains informative when camera-based capture is incomplete.

This work contributes to a growing body of research applying deep neural networks to gait analysis ^{46 12,14}, and strongly underscores the value of this shift by demonstrating predictive capacity across body systems rather than within a single pathology. We acknowledge several limitations. Our cohort is composed predominantly of Ashkenazi Jewish adults residing in Israel. Although the cohort also includes participants of North African, Near Eastern, and other ancestries, generalizability to ethnically and geographically diverse cohorts therefore remains to be determined. To validate the predictive signal is not confounded by ancestry or population we show that gait embeddings predictions transfer across ancestry groups and that removing related-household individuals from evaluation does not reduce gait predictive power (Supplementary Figure 5, 6). Our cohort comprises community-dwelling adults without major diagnosed disease at enrollment. Generalizability to clinical cohorts therefore remains to be established. The model is established for a healthy population, a weakness as much as it is a strength. Prospective validation in clinical settings will be needed to establish diagnostic utility but it shows that gait carries systemic health information beyond pathology-specific dysfunction. The pose-estimation algorithm tied to the Azure Kinect hardware is the hard barrier to proving generalization on public datasets, since estimators differ in joint definitions^{47,48}; training future models on widely used, standardized pose-estimation algorithms will allow cross-system generalizability, although it may degrade predictive performance. All reported associations are cross-sectional and cannot support temporal or causal claims, we report preliminary longitudinal results, limited by small n (Supplementary figure 3). Establishing whether gait predicts future changes in health status, including incident disease and medication initiation, will require the longer follow-up that the HPP's 25-year design will be positioned to provide as we collect more data, and is a natural direction for future work. Future directions include longitudinal prediction of disease onset, external validation on ethnically and geographically diverse cohorts, translation to consumer-grade 2D cameras, upper limb probes ⁴⁹, and the design of targeted motor assessments, such as trunk-focused tasks for sleep and lifestyle screening, or lower-limb wearable protocols for metabolic and frailty markers.

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Methods

Description of cohort

The data presented in this paper were collected between January 2023 and December 2025, from a total of 3,328 participants aged between 20 years and 79 years, who were enrolled as part of the HPP study and who underwent gait activity analysis as part of the study. This study was approved by the institutional review board of the Weizmann Institute of Science (reference no. 1719-1), and all participants were self-assigned volunteers with informed consent. The cohort in this study is one of the largest longitudinal studies established in Israel, with a population originating from several different ancestries who reside in a relatively small geographic region and, therefore, share a relatively similar environment and habits. The population is largely composed of educated European (Ashkenazi) Jews, healthy at the time of recruitment (that is, severe medical conditions and diseases were defined as exclusion criteria), who have follow-up visits every 2 years for 25 years¹⁸. For the full study design, see¹⁷. The data include various clinical, physiological, behavioral and multi-omic profiling data, which we categorized into 18 body systems (excluding gait characteristics) representing major physiological systems and environmental exposures (**Figure 1**).⁵⁰ The following groups represent the baseline characteristics and system-level categories used for this work. The exact number of individuals, data points and main features available in each group can be found in Extended Table 1.

Anthropometric: includes 3 body measurements collected during the clinical visit: waist circumference, hip circumference, and body weight, representing basic anthropometric indices of body size and central adiposity. BMI, height, and detailed regional body composition measurements were excluded from this

category as they were accounted for in the baseline covariate group and body composition system, respectively.

Blood Tests lipids: includes the following blood tests: high-density lipoprotein cholesterol, non-high-density lipoprotein cholesterol and triglycerides.

Body composition: includes 108 measurements of fat and lean mass assessments for the legs, arms, trunk, gynoid and android regions. These measurements are derived from dual-energy X-ray absorptiometry imaging. Additionally, anthropometric parameters, such as weight, height and hip and waist circumference, were also included. BMI was not part of this category because it was already accounted for in the baseline group.

Bone density: includes 182 measurements of the mineral content in different parts of a variety of skeletal components, based on dual-energy X-ray absorptiometry imaging.

Cardiovascular system: includes measurements from various tests: blood pressure measurements, blood pressure ratios computed using the ankle-brachial index test, arterial stiffness estimated by pulse wave velocity using a Falcon device (Viasonix) and carotid intima-media thickness computed from the carotid ultrasound using a SuperSonic Aixplorer MACH 30 (Hologic), vascular parameters averaged over both eyes computed from retinal imaging using an iCare DRSplus confocal fundus imaging system (iCare) and the Python AutoMorph package⁶¹, as well as electric activity of the heart as captured in the 12-lead resting ECG using a PC-ECG 1200 machine (NORAV).

Frailty: includes 6 features capturing the key components of physical frailty: appendicular lean mass of the four limbs (right and left leg, right and left arm) derived from dual-energy X-ray absorptiometry and bilateral hand grip strength measured by dynamometry.

High-level diet: includes 124 measurements encompassing macronutrients, micronutrients, meal patterns, and food categories. This system integrates fundamental nutritional components, including energy intake, proteins, carbohydrates, and dietary fiber - alongside 13 vitamins, 11 minerals, and specific profiles for fatty acids and sugars. Behavioral dietary aspects are captured through 29 distinct food categories (e.g., whole grains, vegetables, and processed meats) and 4 meal pattern metrics such as fasting window and meal frequency. These features are derived from mean daily consumption values calculated from longitudinal diet logging records within a window around each research stage, using a 500 kcal daily energy threshold to ensure data reliability. Finally, macronutrient caloric ratios and lipid distribution metrics were included to provide a normalized assessment of dietary composition for predictive gait modeling.

Gut microbiome: includes relative abundances of 627 families. We used the method for metagenomic reads extraction and bacterial abundances estimation previously described⁵¹ at the family level, in combination with the previously published improved human gut microbiome reference set for mapping⁵². Bacterial families that were not identified in at least 5% of the samples were not included. Missing data were assumed to represent missing abundance or abundances that are below the detection limit. Thus, missing data were imputed by a minimum value of 0.0001. In this work, the log10-transformed values of the resulting abundances per sample were used.

Hematopoietic system: includes the following blood laboratory test measurements, which are characteristics and components of red blood cells: ferritin, hemoglobin, mean corpuscular hemoglobin concentration, mean corpuscular volume, red blood cells, hematocrit, mean corpuscular hemoglobin and red blood cell distribution width. Immune system: includes the following immune cell measurements from complete blood count: white blood cells, absolute eosinophils, percentage eosinophils, absolute

monocytes, percent- age monocytes, absolute lymphocytes, percentage lymphocytes, absolute basophils, percentage basophils, absolute neutrophils and percentage neutrophils.

Glycemic status: includes 49 parameters describing blood glucose variability computed as previously described ⁵³. In addition, this category includes fasting glucose and HbA1C from laboratory tests.

Lifestyle: assessed by 46 questions from questionnaires, following the structure and content of the UK Biobank ⁵⁴, addressing the following topics: smoking, alcohol consumption, physical activity, employment, sleep, sun exposure and electronic device use (**full list in Supplementary Table 1**).

Liver health: assessed by parameters from a liver ultrasound and 2D shear wave elastography (2D-SWE), including measurements of viscosity, elasticity, attenuation and sound speed. In addition, the following liver enzymes and related blood tests are included: alkaline phosphatase, alanine transaminase, aspartate transaminase, total protein, total bilirubin, platelets and albumin.

Nightingale NMR: includes serum metabolic biomarkers quantified by the proton nuclear magnetic resonance (¹H-NMR) platform of Nightingale Health. This platform provides simultaneous quantification of 228 absolute-value-based plasma metabolites and ratios, mainly expanding detailed lipidomic profiles and adding measurements of clinically validated biomarkers, including routine lipids, lipoprotein subclass profiling with lipid concentrations within 14 subclasses, fatty-acid composition, and various low-molecular-weight metabolites such as amino acids, ketone bodies, and glycolysis metabolites ⁵⁵.

Serum untargeted metabolomics (n = 1X,000) was performed via UPLC-IMS-MS (Waters SELECT SERIES Cyclic IMS) in batches of 416 injections. Processed/solvent blanks were used for background subtraction, while pooled QCs (intra-batch) and "Anchor" samples (inter-batch) corrected for analytical drift. Post-alignment, 1,809 high-quality features were retained. Three gait-associated leads, including LysoPE(P-16:0/0:0), achieved MSI Level 2 status (< 0.5 ppm mass accuracy, < 1% CCS error, and MS/MS fragments). Remaining features, such as a C₁₇ oxylipin-like series (C17H24O4-5), were assigned MSI Level 4 formulas via < 0.8 ppm accuracy and > 90% isotopic match.

Sleep: includes 448 characteristics derived from multi-night home sleep apnea testing using the WatchPAT device (ZOLL Itamar) ¹⁸. Features encompass respiratory indices (apnea-hypopnea index, oxygen desaturation index, respiratory disturbance index), sleep architecture parameters (total sleep time, sleep efficiency, REM and non-REM durations), body position, snoring metrics, and heart rate dynamics during sleep.

Mental health: assessed by 35 questions from a questionnaire following the structure and content of the UK Biobank ⁵⁴ equivalent questionnaire relating to the individual's mood, satisfaction and depressive symptoms. In addition, recent depressive symptoms (RDS) score was computed as the sum of the self-reported scores of the following questions; each self-reported score is a number between 1 and 4 (1 indicates not at all, 4 indicates nearly every day) and, therefore, ranges from 4 to 16: Over the past two weeks, how often have you felt down, depressed or hopeless? Over the past two weeks, how often have you had little interest or pleasure in doing things? Over the past two weeks, how often have you felt tense, fidgety or restless? Over the past two weeks, how often have you felt tired or had little energy? The list of the 36 total features for this body system can be found in **Supplementary Table 2**.

Renal function: includes creatinine, urea and the electrolytes sodium and potassium, all derived from blood laboratory tests.

Medications: includes a total of 60 medication types, with self-reports relating to if the medication was taken between the study enrollment and the data point or not (1 or 0, respectively).

Medical conditions: 127 self reported medical conditions.

Data Acquisition and Preliminary Filtering

Participants completed a protocol of up to six movement assessment activities, captured using the Movement Lab (Newton VR Ltd., hereafter Newton Tech): treadmill walking at a fixed speed of 3 km/h (3 minutes), treadmill walking at a self-selected pace (1 minute), a 30-second sit-to-stand test, stationary walking (1 minute), and Romberg test with eyes closed for 30 seconds. Not all participants completed every activity; data gaps resulted from technical issues (data corruption) or safety protocols (e.g., exclusion of treadmill tasks for participants lacking appropriate footwear). The protocol was designed to probe complementary aspects of locomotor function, postural control, and functional capacity, with a total administration time of approximately 10 minutes per participant. We use 'gait' throughout to denote this full protocol and not treadmill locomotion alone; task-specific results are identified by the relevant activity where relevant.

The Movement Lab was used in this instance with the Azure Kinect Body Tracking SDK (K4ABT), which provides 3D spatial coordinates (x, y, z) and confidence scores for 32 anatomical joints at a sampling rate of 30 frames per second (fps) (see **Extended Figure 7**). All recordings were performed from a frontal camera perspective. To ensure data fidelity we removed six distal joints with high variance and low detection confidence (HAND_LEFT, HANDTIP_LEFT, THUMB_LEFT, HAND_RIGHT, HANDTIP_RIGHT, and THUMB_RIGHT), retaining a 26-joint skeleton for analysis. These joints were identified empirically as having consistently low confidence scores, (i.e. <10% of joints were above confidence 0) across the dataset.

We utilized a preprocessing pipeline designed to preserve the raw temporal dynamics of gait. Input sequences were fixed at 900 frames (~30 seconds at 30 fps) as most of the activities were 30 seconds long. All motor activities were designed to last at least 30 seconds, so no padding was required for valid recordings. Trials shorter than 30 seconds resulted solely from acquisition errors and accounted for fewer than 0.2% of all trials; these were excluded rather than padded or truncated. To eliminate high-frequency shot noise, a temporal median filter with a window size of 3 frames was applied to each joint coordinate independently using a nearest-neighbor boundary mode. Spatial normalization was performed on a per-frame basis. Each skeleton was centered by subtracting the centroid computed from the pelvis joint (or pelvis and spine navel joints, depending on configuration). Following centering, the coordinates were scaled by the furthest joint distance from the origin within that frame, computed as the maximum Euclidean norm across all joints. This approach ensures that the spatial representation is invariant to the subject's absolute position and body size while strictly maintaining the original temporal structure and cadence of the walking sequence.

Gait Feature Extraction

Detailed gait analysis was performed on the 3D skeletal sequences by Newton Tech's Movement Lab - a commercial motion analysis pipeline. For walking tasks (treadmill and stationary walking), spatiotemporal parameters included cadence, bilateral step and stride length, step width, walking speed, swing and stance phase durations, double support time, and bilateral asymmetry indices for step length,

step time, swing, and stance. Upper-body dynamics were captured through bilateral arm swing amplitude and head vertical sway. For postural tasks (stationary walking and Romberg test), center-of-mass sway was quantified using displacement metrics (anteroposterior and mediolateral distance, RMS distance, 95% confidence ellipse area), velocity metrics (mean and peak sway velocity), frequency-domain measures (mean frequency, total power, centroidal frequency, frequency dispersion), and derived indices including sway amplitude, normalized jerk, and sway path²⁶. For the sit-to-stand task, extracted features included repetition count, chair-rise time, chair-rise time variability, peak reaction force normalized to body weight, and maximal rate of force development. In total, 71 unique parameter definitions computed across the five motor tasks yielded 672 activity-specific features. To reduce redundancy, we computed pairwise Pearson correlations across all features and applied hierarchical clustering with a correlation threshold of $|r| > 0.85$; a single representative feature was selected from each cluster, yielding 139 non-redundant descriptors. This target-agnostic reduction preserves the information content of the original feature set while eliminating collinear redundancy. These descriptors were combined with participant height to form the engineered baseline feature set used throughout this study (Full feature list and selected features are in **Supplementary Table 3**)

Deep Learning Architecture and Pretraining

The gait representation model is based on a Dual-Stream Spatio-Temporal Transformer (DSTformer) architecture (**Supplementary Figure 7**), inspired by the MotionBERT framework²⁷. The model processes four-channel gait sequences (X, Y, Z, C), where the first three channels represent 3D joint coordinates and the fourth channel provides a per-joint detection confidence score from the pose estimation system. To capture long-range gait dynamics, the model was configured to process 900-frame trajectories (30 seconds of continuous movement at 30 fps). This was made computationally efficient through the integration of Rotary Position Embeddings (RoPE)⁵⁶ and the use of Flash Attention (via PyTorch's `scaled_dot_product_attention`)⁵⁷, in the temporal attention blocks enabling sequence length extrapolation without fixed positional embeddings, as well as being a better architecture for long sequence which achieved $O(n)$ memory complexity through chunked attention computation. The architecture consists of 8 transformer blocks with a feature dimension of 128 and 8 attention heads. Each block features parallel spatial-temporal (ST) and temporal-spatial (TS) attention streams that are integrated via learned fusion weights, allowing the model to adaptively balance the contribution of spatial joint relationships and temporal movement dynamics. The spatial attention operates across joints within each frame with absolute Positional Encoding, while the temporal attention operates across frames for each joint, with RoPE applied only to the temporal dimension to encode relative position information. For sequence reconstruction, we employed a decoder consisting of 3 Linear Layer and Tanh activation that maps the latent embeddings back to the original 3D coordinate space.

Data Augmentation

To enhance the model's generalization capabilities and account for sensor noise and natural movement variability, we implemented a data augmentation strategy using Gaussian jittering. Additive white Gaussian noise was applied to the joint coordinates, sampled from a normal distribution with a standard deviation of $\sigma=0.05$ (in normalized coordinate space). This augmentation effectively simulates the inherent precision limitations of the pose-estimation system without distorting the underlying gait patterns. A hybrid training protocol was employed wherein the model was exposed to both the original (clean) and the augmented (noisy) sequences. This dual-exposure strategy allowed the network to simultaneously learn noise-robust features and high-fidelity motion reconstruction.

Pretraining

Pretraining was conducted using a Denoising Masked Autoencoder (DMAE) objective. In this setup, joint positions were stochastically masked by zeroing out their input channels according to the multi-level masking strategy described above, forcing the model to reconstruct the missing spatio-temporal information. When adding noise augmentation we added 0.05 units of gaussian noise. The optimization objective combined Euclidean distance reconstruction loss (MPJPE) with a 3D velocity loss with $\lambda = 0.1$ that penalizes discrepancies in frame-to-frame joint velocities, and scaled Mean Per-Joint Position Error (NMPJPE) with $\lambda = 0.1$ to ensure anatomically consistent reconstructions.

$$Loss = MPJPE + \lambda_1 NMPJPE + \lambda_2 VelocityError$$

All models were trained on 1 NVIDIA L40S GPU using PyTorch⁵⁸ with the AdamW optimizer⁵⁹ (weight decay 0.01), a cosine annealing learning rate schedule⁶⁰, and linear warmup over the first 0.3 epochs. To maximize hardware utilization, we utilized mixed-precision training⁶¹ and torch.compile graph optimization for accelerated computation. Pretraining used all available recordings. Recordings from a preliminary room were excluded from downstream prediction to reduce batch effects. Pretraining on the combined set outperformed pretraining on the main-cohort recordings alone. No participant recorded in the preliminary room entered any cross-validation fold. We reserved 10% of the data for evaluation; the model achieved nearly identical reconstruction performance on this held-out set compared to the training set, demonstrating strong generalization and the successful preservation of high-fidelity gait characteristics within the learned embedding space.

Masking Strategy

The model was pretrained using a Masked Autoencoder (MAE) framework with a composite masking strategy designed to achieve a high effective masking ratio and encourage the learning of robust structural and temporal embeddings. The masking strategy combined:

1. Anatomical Group Masking: Joints were partitioned into six anatomical regions: Left Leg (hip, knee, ankle, foot), Right Leg, Left Arm (clavicle, shoulder, elbow, wrist), Right Arm, Torso (pelvis, spine navel, spine chest, neck), and Head (head, nose, eyes, ears). During training, 4 out of 6 regions were randomly selected and masked across temporal spans of 16 consecutive frames.

This spatial-temporal span masking forced the model to reconstruct limb-specific trajectories based on the global coordination of the remaining body segments.

2. **Random Frame Masking:** Entire frames were randomly selected and masked with a probability of 5% across the sequence, forcing the model to interpolate missing temporal information from preceding and succeeding movement states.

To evaluate the effectiveness of each masking strategy, we trained a model with each component ablated and compared age prediction performance detailed in Extended Table 2. The full strategy (anatomical group masking plus random frame masking) achieved the highest performance (male Pearson $r = 0.702$; female $r = 0.676$). Removing frame masking (male $r = 0.682$; female $r = 0.673$) or group masking (male $r = 0.671$; female $r = 0.656$) reduced accuracy, and noise augmentation alone performed worst (male $r = 0.665$; female $r = 0.635$). Group masking contributed more than frame masking, and the two were complementary. Removing Gaussian noise augmentation also reduced performance (male $r = 0.631$; female $r = 0.599$). Data were collected over a period of more than two years during which system calibration varied across sessions, as reported by the acquisition team; noise augmentation exposes the model to a controlled distribution of such perturbations.

The combination of these masking strategies results in an effective masking ratio of approximately 70% of the input signal, creating a challenging reconstruction objective that promotes the learning of rich, generalizable representations.

Hierarchical Pooling and Feature Aggregation

To transform variable-length kinematic sequences into standardized physiological representations, we benchmarked five hierarchical pooling architectures on age prediction with 5-fold cross validation. The objective was to identify an aggregation strategy that balances the preservation of anatomical structure with robustness to temporal variability. We evaluated strategies ranging from baseline global statistics to fine-grained joint-specific embeddings. As detailed in **Extended Table 3**, the analysis indicated that increasing spatial granularity does not strictly correlate with predictive performance. For instance, maintaining independent representations for all 26 joints (Variant 4) or strictly separating bilateral limbs (Variant 3) resulted in lower correlations, likely due to overfitting local noise or sensitivity to minor gait asymmetries. The selected architecture, Variant 5 (Bilaterally-Merged Groups with Percentile Pooling), demonstrated the highest Pearson correlation ($r=0.5765$). This strategy aggregates skeletal data into four functional domains- Head, Torso, Arms, and Legs- by merging bilateral limb data to reduce asymmetry-induced artifacts. Furthermore, by utilizing the 99th percentile alongside the temporal mean, Variant 5 captures both steady-state dynamics and transient peak activations (e.g., heel-strike impact), yielding a robust 1024-dimensional physiological signature.

Downstream Evaluation Framework and Statistical Analysis

Nested Cross-Validation Strategy. To ensure rigorous evaluation and account for the longitudinal structure of the dataset, we implemented a nested, subject-level cross-validation (CV) framework. Data

splitting was performed strictly at the subject level to prevent data leakage between visits. We employed a 5-fold outer loop for performance estimation, repeated across 15 distinct random seeds (generating $n=75$ independent train-test splits) to produce stable mean estimates and confidence intervals. For binary targets with significant class imbalance ($<20\%$ minority class), stratified splitting was applied to maintain consistent class distributions.

Model Selection and Hyperparameter Optimization. Within the inner CV loop (4-fold), we decoupled model selection from performance evaluation. We competitively evaluated linear models (Ridge regression for continuous targets; Logistic regression for binary/ordinal targets) against nonlinear gradient-boosted decision trees (LightGBM)⁶². Hyperparameters (e.g., regularization strength, tree depth, learning rate) were optimized via RandomizedSearchCV. The optimal model architecture was selected automatically based on inner-loop performance, ensuring that the outer-loop test data remained unseen during the optimization process.

Ensemble Architecture. To integrate evidence from diverse movement contexts, we utilized a unified ensemble approach. The feature matrix was constructed by concatenating the 1,024-dimensional latent embeddings with activity-specific metadata (activity type and sequence index). Categorical metadata were one-hot encoded for linear models and ordinally encoded for tree-based models. Final predictions were generated using a late-fusion strategy: model outputs were averaged across all gait activities per subject-visit pair to produce a single consensus prediction.

Direct Prediction Analysis. To evaluate the raw predictive capacity of gait embeddings without demographic or anthropometric adjustment, we trained models using gait embeddings as the sole input features. For each of the 3,210 phenotypic targets, predictions were evaluated using 5-fold nested cross-validation repeated across 15 random seeds. Statistical significance was assessed using the Pearson correlation p-value for each target, with correction for multiple comparisons using the Benjamini–Hochberg FDR procedure at $q < 0.05$. This analysis simulates a passive screening scenario in which only a gait recording is available, with no accompanying clinical information (Figures 3, Extended Figure 1).

Covariates and Baseline Definitions. To isolate the specific predictive utility of gait physiology beyond standard demographic and anthropometric factors, we utilized Age, Sex, Body Mass Index (BMI), and Visceral Adipose Tissue (VAT) as covariates. For all predictive tasks, we constructed a null baseline model trained solely on these covariates. This design allows for the explicit quantification of the added information value provided by the gait embeddings.

Recording duration and intermittent sampling analysis. To assess data requirements, we evaluated prediction performance as a function of recording length using treadmill walking at 3 km/h. We extracted recordings of 2, 5, 10, 15, 30, and 180 seconds and re-extracted embeddings for each length. To assess intermittent capture, we assembled recordings of fixed total length from non-adjacent short clips drawn at random from the full 180 second walk, under three protocols (5×2 s, 6×5 s, 2×15 s). All models used gait embeddings combined with height and were evaluated with the same nested cross-validation procedure and fold assignments described above. Performance was reported as Pearson r between predicted and measured values.

Normalized Keypoint Baseline. We evaluated a baseline in which the normalized 3D skeletal trajectories were used directly as regression inputs. These trajectories were produced by the same centroid-based spatial normalization and the same long-sequence temporal sampling (900 frames per recording, 26 body keypoints, x/y/z coordinates). Because the full flattened trajectory ($900 \times 26 \times 3 \approx 70,200$ features per sequence) is too high-dimensional for downstream regression, we summarized each recording with a compact temporal pooling: the 900 frames were partitioned into 30 non-overlapping 30-frame blocks, and within each block we computed the mean and maximum of every joint-axis channel, yielding a 4,680-dimensional feature vector ($30 \text{ blocks} \times 26 \text{ joints} \times 3 \text{ axes} \times 2 \text{ statistics}$). These pooled feature vectors were then passed through the downstream evaluation pipeline used for the gait embeddings.

Lower-limb ablation model. A separate model was pretrained on the nine lower-limb joints: bilateral hips, knees, ankles, feet, and the pelvis. The architecture and training objective matched the full-body model. Random joint masking and random frame masking as well as noise augmentation was applied within the reduced joint set. Embeddings were extracted and passed through the downstream evaluation pipeline and fold assignments used for the full-body model.

Inference and Statistical Validation. The final predictive power for each model was reported as the median metric (Pearson r for regression; AUC-ROC for classification) across the seeds of predicted folds. To rigorously validate associations, we compared the predictive performance of the full model (Gait + Covariates) against the baseline null model. For each target, we performed a two-sided t-test comparing the distribution of scores from the full model ($n=15$) against the distribution of scores from the baseline model ($n=15$). A gait association was considered statistically significant only if the full model's performance distribution was significantly higher than that of the baseline after BH-FDR correction ($\text{FDR } q < 0.1$).

Evaluation Metrics. Performance was assessed using metrics appropriate for the statistical properties of each biomarker: Pearson correlation coefficient (r) and coefficient of determination (R^2) for continuous regression targets; Area Under the Receiver Operating Characteristic Curve (AUC-ROC) for binary classification; and Spearman rank correlation (ρ) for ordinal variables. Given the high discriminative power of gait features for sex ($\text{AUC} \approx 0.99$), analyses were performed on both the full cohort and stratified by sex to detect sex-specific physiological associations.

Liver Blood Tests Baseline. The liver blood tests baseline in Extended Figure 5 is the liver blood tests markers: alkaline phosphatase, alanine transaminase, aspartate transaminase, total protein, total bilirubin, platelets and albumin added to the demographics age and sex passed through the downstream evaluation pipeline predicting liver elasticity.

Association with current diseases and medical conditions To identify associations between gait embeddings and diseases, medical conditions or medication intake, we applied a logistic regression model with a ridge regularization (using the scikit-learn ⁶³ Python package), for male and female participants separately. Inference and validation were performed as mentioned above, with the exception of the model score calculation, which was computed as the AUC for each iteration, to adapt to binary classification.

Translating AUC into clinical referral. To express discrimination gains as screening yield, we ranked individuals by predicted probability and set the operating point at the top 10%. We report all conditions with a significant ΔAUC for the gait-plus-Age+BMI model over the Age+BMI baseline. For each, we report sensitivity at the top-10% referral threshold, defined as $\text{TP}/(\text{TP}+\text{FN})$: the fraction of true cases falling within the referred group. Sensitivity was computed per seed and averaged. The gain over baseline was tested by paired wilcox test across seeds and FDR-corrected within sex. Endpoints significant at $q < 0.1$ are marked with an asterisk.

Multiple Testing Correction. To address the multiple comparisons problem inherent in high-dimensional biomarker analysis, statistical significance was determined using the Benjamini-Hochberg False Discovery Rate (FDR) procedure. Correction was applied independently for each feature group (hypothesis family).⁶⁴

Anatomical Interpretability and Feature Attribution

To decipher which biomechanical components drive specific physiological predictions, we developed a post-hoc interpretability framework based on systematic joint perturbation. This analysis was conducted on a frozen backbone.

Ablation analyses were performed exclusively on embeddings derived from treadmill walking activities (3 km/h and self-selected pace). Activities with fundamentally different biomechanical demands (sit-to-stand, Romberg test, stationary walking) were excluded from this analysis, as the relative contribution of each joint group to task execution differs inherently across motor paradigms, confounding the attribution of importance to health-related signals.⁶⁵

Anatomical Partitioning We partitioned the 26-joint skeleton into four functional domains based on anatomical adjacency:

- **Head:** Joints 20–25 (Head, nose, eyes, ears).
- **Torso:** Joints 0–3 (Pelvis, spine navel, spine chest, neck).
- **Arms:** Joints 4–11 (Bilateral clavicles, shoulders, elbows, wrists).
- **Legs:** Joints 12–19 (Bilateral hips, knees, ankles, feet).

Perturbation Strategy For each target biomarker and each anatomical group g , we computed performance metrics under two distinct masking conditions:

1. Masked Drop (Δ_{drop}): The joints in group g were zeroed out while retaining the rest of the body. This measures the *necessity* of the region (i.e., how much performance degrades in its absence).

$$\Delta_{drop} = S_{baseline} - S_{(Masked)}$$

2. Isolation Score (r_{solo}): All joints *except* those in group g were zeroed out ("All-But" strategy). This measures the *sufficiency* of the region (i.e., how much predictive signal the region contains independently).

$$r_{solo} = S_{(Isolated)}$$

Importance Calculation A direct summation of these metrics is problematic because Δ_{drop} represents small relative deltas (typically -0.05 to 0.17), whereas r_{solo} represents absolute performance scores (typically 0.1 to 0.7). To prevent the magnitude of the isolation score from dominating the attribution, we implemented a dual-normalization strategy.

Let D be the set of drop scores for all four groups and R be the set of isolation scores. We first Min-Max normalized each component independently across the four groups:

$$\hat{\Delta}_{drop} = (\Delta_{drop} - \min(D)) / (\max(D) - \min(D)) \quad \hat{r}_{solo} = (R_{solo} - \min(R)) / (\max(R) - \min(R))$$

The final importance score I_g was computed as the unweighted sum of these normalized components, followed by a final scaling to the unit interval [0, 1] for visualization:

$$I_g = \text{Norm}(\hat{\Delta}_{drop} + \hat{r}_{solo})$$

This composite metric ensures that a body region is deemed "important" only if it is both necessary for peak performance and sufficient to drive prediction independently.

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Data Availability

Data in this paper are part of the Human Phenotype Project (HPP) and are accessible to researchers from universities and other research institutions at <https://humanphenotypeproject.org/data-access>.

The HPP data include personal information and, in compliance with institutional review board regulations, cannot be made publicly available. Interested bona fide researchers should contact info@pheno.ai to obtain instructions for accessing the data, which is typically granted within a few days.

Code Availability

Code used in this study will be available at <https://github.com/AdamGabet/GaitPredict> (access available upon request during review).

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Contributions

A.G. conceived the project, designed the DL model and conducted all analyses, interpreted the results and wrote the manuscript. S.K., G.L., G.S and A.Z. reviewed the results and the manuscript. S.G. interpreted the results and reviewed the manuscript. A.W. developed protocols and oversaw sample collection and processing. Y.B., A.R. and O.D. collected the data and reviewed the manuscript. N.G. and D.K. Performed data preprocessing. R.S. collected and annotated data. E.S. conceived and directed the project and analyses, designed the analyses and reviewed the results and the manuscript.

Competing Interests

A.R., O.D., and Y.B. are employees of Newton VR Ltd., which developed the Movement Lab system used for gait data acquisition in this study. A.R. is also affiliated with BioPilot AI. A.W. and E.S. are paid consultants of Pheno.AI, Ltd. All other authors declare no competing interests.

Extended Data

Extended Table 1

	Demographics		Treadmill walking 3KPH (5400s)		Stationary Walk (1800s)		Treadmill Walking at Self Selected Pace (1800s)		Romberg's test eyes closed (900s)		Sit to Stand (900s)	
Characteristic, mean (std)	Age	BMI	Cadence (steps/min)	Stride Length (m)	95% Ellipse Area (mm ²)	Sway Amplitude AP (mm)	Cadence (steps/min)	Stride Length (m)	95% Ellipse Area (mm ²)	Sway Amplitude AP (mm)	Repetitions(count)	Chair-Rise Time (ms)
All (3,328)	52.48 (10.10)	25.74 (3.79)	97.66 (10.24)	1.01 (0.11)	62.28 (38.09)	26.17 (8.92)	100.53 (12.36)	1.04 (0.19)	0.36 (0.34)	294.42 (106.82)	9.87 (2.33)	684.12 (110.3)
Male (1,607)	52.07 (9.98)	26.18 (3.28)	94.41 (8.40)	1.06 (0.09)	62.24 (37.36)	24.93 (8.41)	99.02 (11.74)	1.11 (0.18)	0.38 (0.39)	307.06 (98.82)	9.72 (2.36)	715.6 (111.7)
Female (1,721)	52.87 (10.21)	25.34 (4.17)	100.76 (10.45)	0.99 (0.10)	64.17 (39.94)	27.36 (9.23)	101.67 (12.58)	0.99 (0.17)	0.33 (0.29)	282.22 (99.83)	10.01 (2.29)	664.1 (99.9)

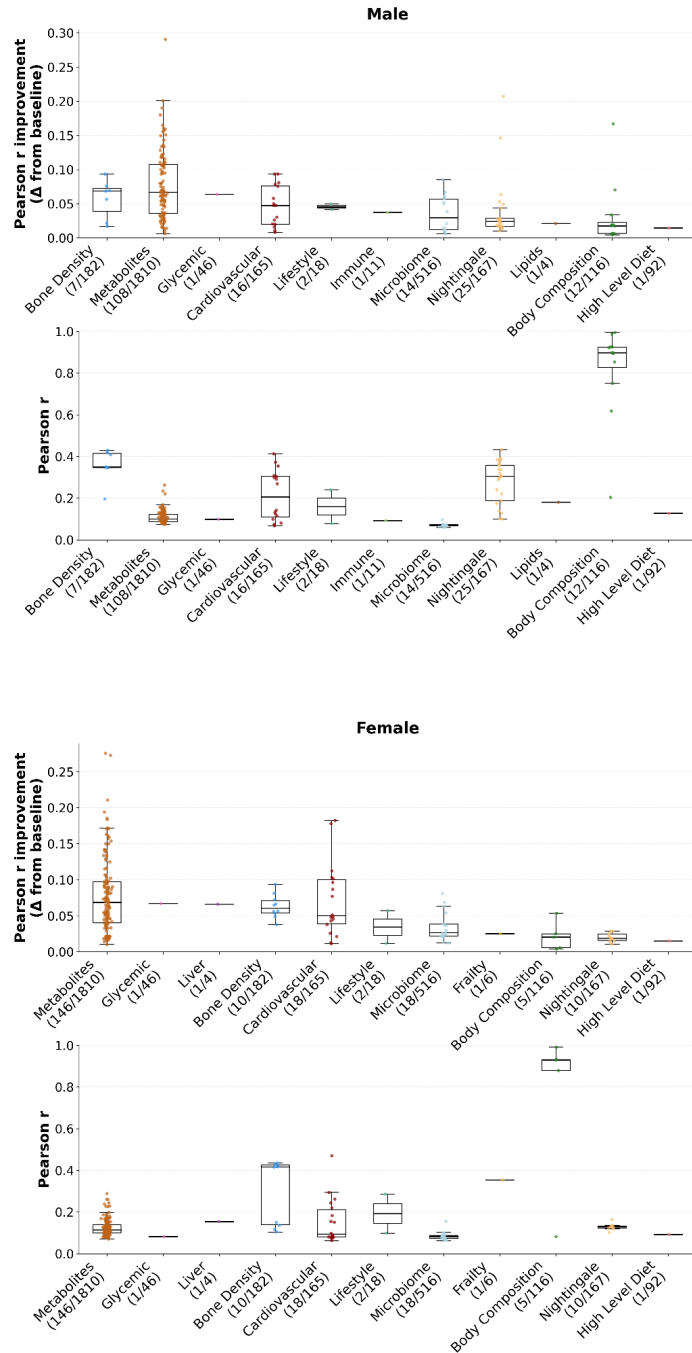
Extended Table 2

Sex	Masking strategy	Age Prediction (Pearson r)
Male	Full masking: frame + group	0.702
	No frame masking	0.682
	No group masking	0.671
	Full masking: no noise	0.631
	Noise only; no frame/group masking	0.665
Female	Full masking: frame + group	0.676
	No frame masking	0.673
	No group masking	0.656

	Full masking: no noise	0.599
	Noise only; no frame/group masking	0.635

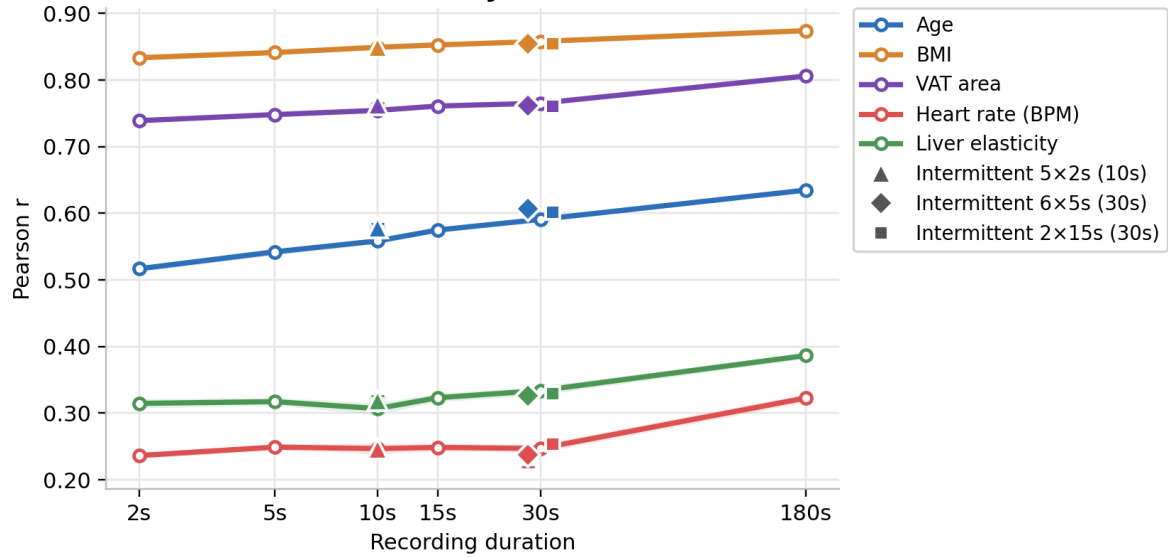
Extended Table 3

Pooling Strategy	Feature Dim.	Age prediction (Pearson r)	Description & Rationale
V1: Mean-Max Global	256	0.618	Baseline. Flattens spatio-temporal features using global mean and max. Computationally efficient but loses structural hierarchy.
V2: Hierarchical Stats	384	0.64	Two-stage. 1. Pooling joints with mean, max, std then mean pooling across time.
V3: Anatomical Groups	768	0.659	Pool with mean across time and joints within 6 regions, maintaining separate left/right limb representations.
V4: Per-Joint Pre-logits	832	0.605	Independent pooling for all 26 joints using embeddings after final MLP before reconstruction layer. Lowest performance suggests overfitting to local noise.
V5: Merged Groups + %	1024	0.691	Selected Method. Pool with mean and 99th percentile across time and then 4 joint groups. Merges bilateral limbs to reduce noise.

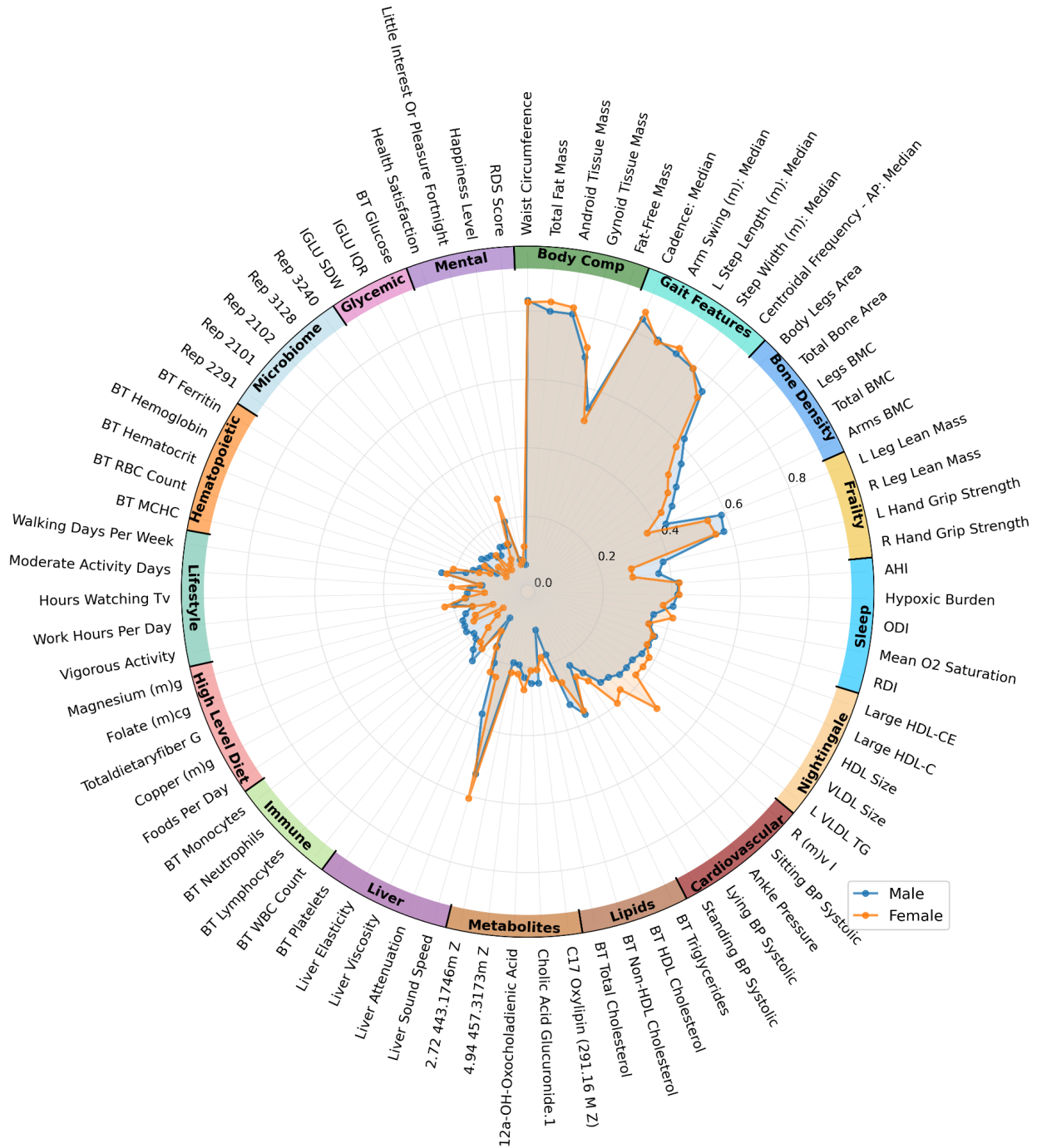


Extended Figure 1: Traditional gait descriptors provide limited predictive gain beyond demographic and anthropometric baselines. Box and swarm plots showing the predictive performance of engineered gait features (cadence, stride length, sway metrics, and height) beyond a demographic baseline (age, BMI, VAT, and height) for male (top) and female (bottom) participants. Top panels show the predictive gain (Δr) attributable to engineered gait features; bottom panels show the median Pearson r of the combined model. Only body systems containing at least one significantly improved feature are

shown. Of 3,210 sex-stratified phenotype predictions tested, 188 were significant in males and 213 in females (FDR-corrected $P < 0.1$), compared with 823 and 723 respectively for learned gait embeddings (Figure 4). Predictions were evaluated using 5-fold nested cross-validation repeated across 15 random seeds ($n = 75$ train-test splits). Significance was assessed by two-sided t-test comparing full versus baseline model performance distributions.

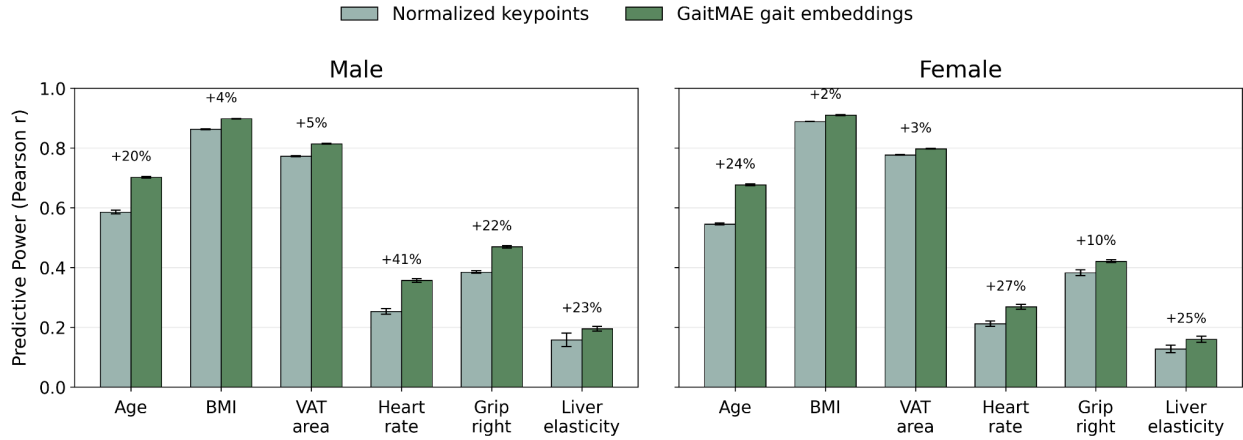


Extended Figure 2: Predictive performance scales with recording length and is robust to intermittent sampling. Pearson r between predicted and measured values as a function of treadmill walking (3 km h^{-1}) recording duration, from 2 to 180 seconds, for five targets spanning morphological and dynamic phenotypes: BMI, visceral adipose tissue (VAT) area, chronological age, heart rate, and liver elasticity. Open circles denote continuous recordings of the indicated length. Filled markers denote intermittent protocols in which the same total recording time was assembled from non-adjacent short clips drawn at random from the full 180 second walk, using a fixed seed per subject: triangle, $5 \times 2 \text{ s}$ (10 s total); diamond, $6 \times 5 \text{ s}$ (30 s total); square, $2 \times 15 \text{ s}$ (30 s total). All models used gait embeddings combined with height, evaluated via 5-fold nested cross-validation repeated across 15 random seeds ($n = 75$ train-test splits) using the same fold assignments as the main analyses. Shaded bands and whiskers denote $\pm$ s.d. across seeds. Body-composition targets approach asymptotic performance within the shortest windows, whereas dynamic targets continue to improve with duration. Across all targets, intermittent sampling matches continuous recording of equal total length. BMI, body mass index; VAT, visceral adipose tissue.



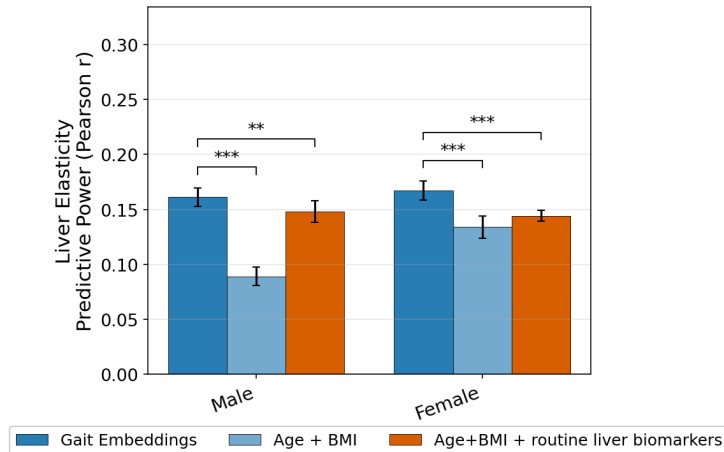
Extended Figure 3: Sex-stratified gait embedding predictions across body systems. Radar plot showing the top 5 features most strongly predicted by the Gait Fusion model for each body system, stratified by sex (male, blue; female, orange). Each slice represents a single phenotype, grouped by body system (labeled in the inner ring). The radial axis spans Pearson r correlation of predictions from 0 to 1. Features within each body system are ordered clockwise. Predictive performance is largely concordant between sexes for structural and metabolic targets (body composition, bone density, lipids), while greater divergence is evident for hematopoietic, lifestyle, and sleep phenotypes. Predictions were generated using linear ridge regression on gait embeddings from the frozen Gait Foundation Model encoder, evaluated via

5-fold nested cross-validation repeated across 15 random seeds ($n = 75$ train–test splits). Of the 3,210 sex-stratified phenotype predictions tested, 959 were significant in males and 1,073 in females after FDR correction. No demographic or anthropometric covariates were included, simulating a passive screening scenario in which only a gait recording is available. AHI, apnea–hypopnea index; BMC, bone mineral content; BT, blood test; HDL-C, high-density lipoprotein cholesterol; HDL-CE, high-density lipoprotein cholesterol ester; IGLU, indices of glycemic variability; MCHC, mean corpuscular hemoglobin concentration; ODI, oxygen desaturation index; RBC, red blood cells; RDI, respiratory disturbance index; VLDL, very-low-density lipoprotein; WBC, white blood cells.



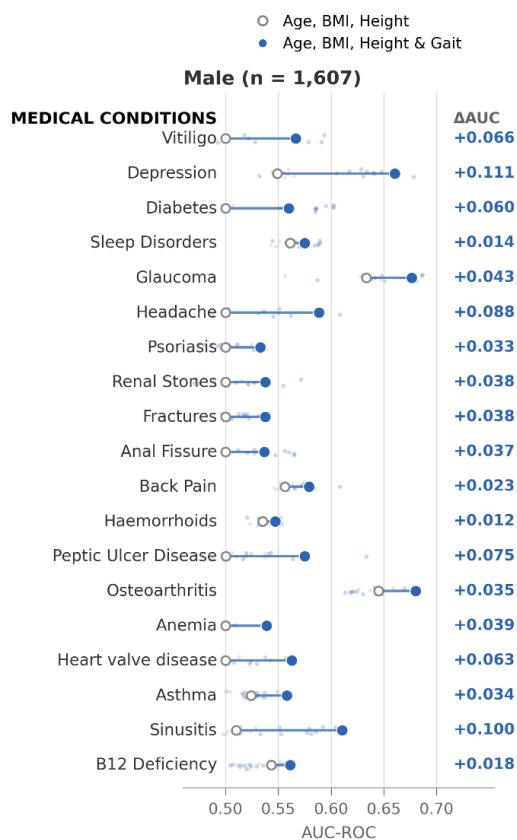
Extended Figure 4: Self-supervised pretraining improves on a normalized-keypoint baseline.

Predictive performance (Pearson r) of a normalized-keypoint baseline (light) versus Gait Foundation Model embeddings (dark) for male (left) and female (right) participants across six representative targets. Keypoint features are the per-joint mean and maximum pooled over non-overlapping 30-frame windows ($26 \text{ joints} \times 30 \text{ windows per } 30\text{-s sequence}$), passed to linear and non-linear models and the same Gait Fusion ensemble used for the learned embeddings. Bars show the mean across 5 seeds (error bars, s.d.). Annotations give the relative gain of embeddings over the baseline ($\Delta r / \text{baseline } r, \%$). The baseline already predicts body-composition targets well (BMI, VAT area); the largest relative gains appear for targets with no trivial anthropometric correlation (age, heart rate, liver elasticity).

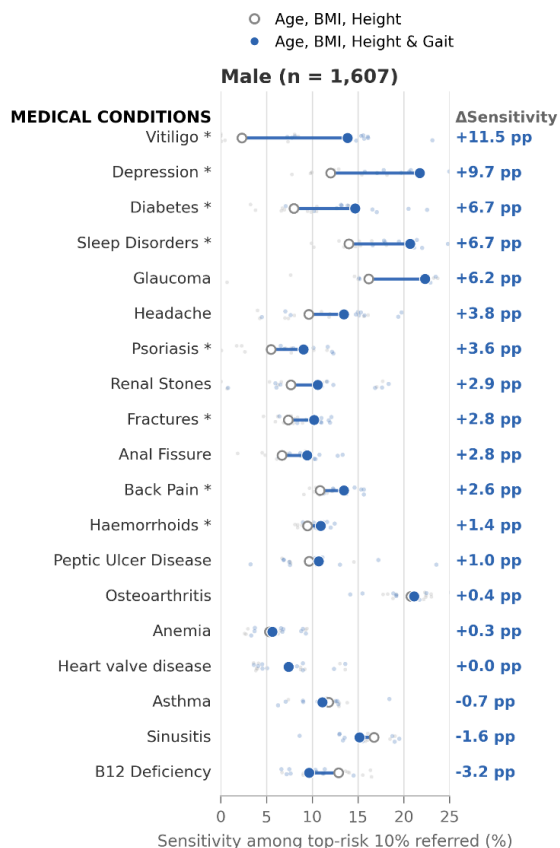


Extended Figure 5 | Gait embeddings predict liver elasticity beyond routine clinical liver biomarkers. Predictive performance (Pearson r) for liver elasticity across three input sets, shown for male and female participants. Gait Embeddings (dark blue) are gait-only predictions. Age + BMI (light blue) is a demographic and anthropometric baseline. Age + BMI + routine liver biomarkers (orange) adds the standard serum liver panel (alkaline phosphatase, alanine transaminase, aspartate transaminase, total protein, total bilirubin, platelets, albumin) to that baseline. Bars show the mean across 15 seeds; error bars denote s.d. Predictions were evaluated using 5-fold nested cross-validation repeated across 15 random seeds ($n = 75$ train-test splits), sex-stratified throughout. Significance was assessed by two-sided t-test comparing performance distributions (** $P < 0.01$, *** $P < 0.001$, FDR-corrected). Gait embeddings alone matched or exceeded a baseline that includes routine clinical liver markers in both sexes.

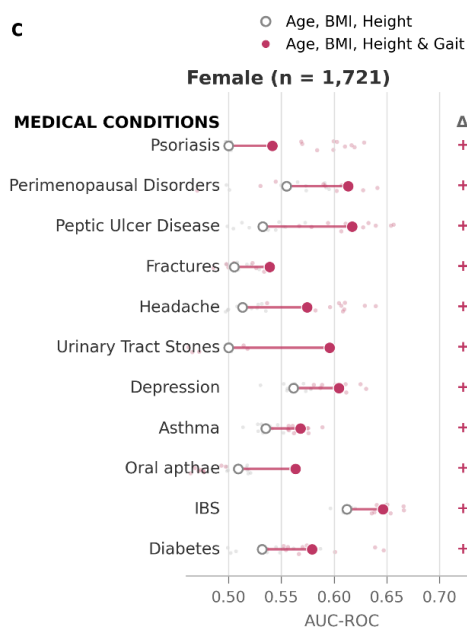
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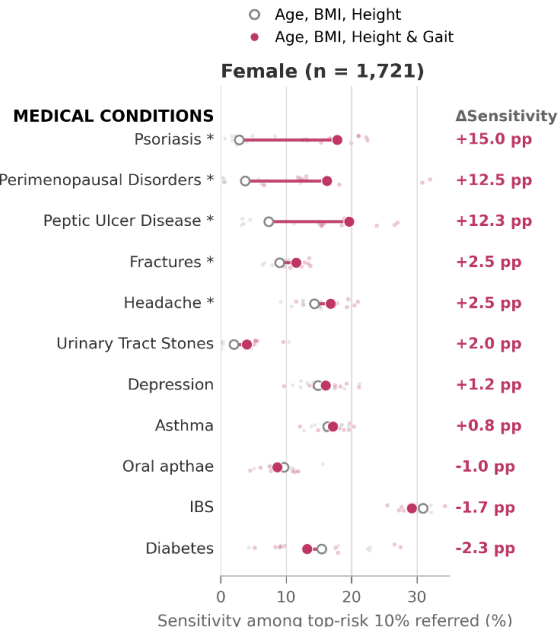
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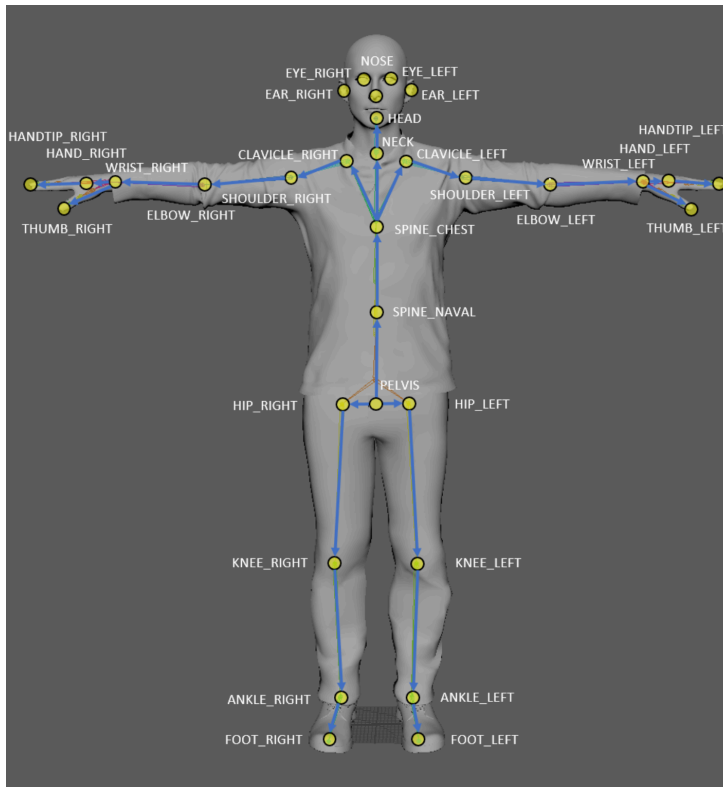


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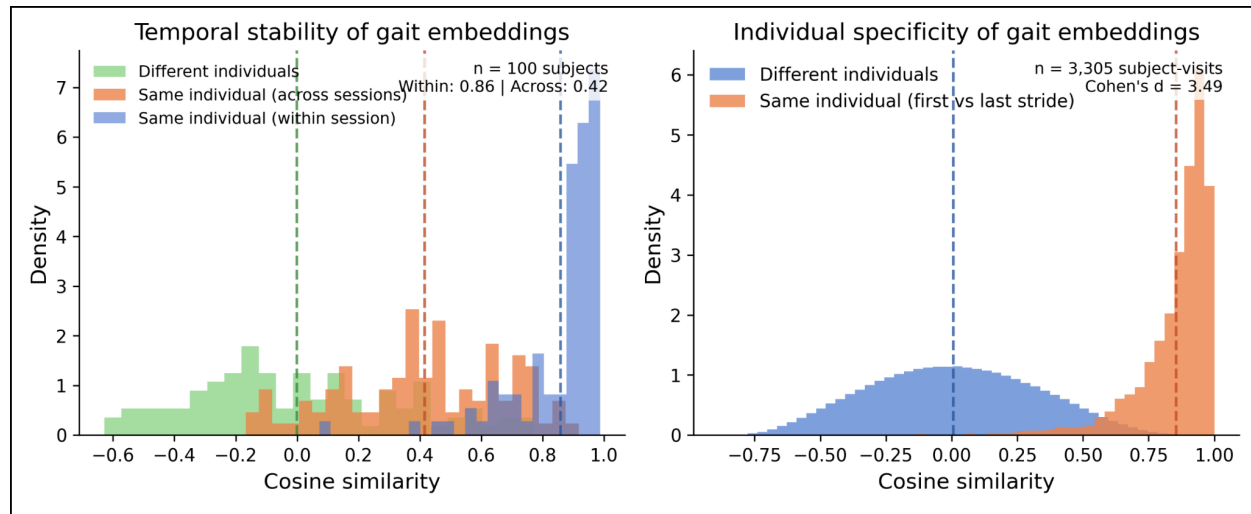
Extended Figure 6 | Translating gait predictions into screening yield: incremental case detection at a fixed referral capacity. Discriminative performance for FDR-significant self-reported conditions, comparing a covariate-only model (age, BMI, height; open circles)

against the same covariates plus gait embeddings (filled circles). **a**, AUC-ROC, males ($n = 1,607$). **b**, Sensitivity among the top-risk 10% referred, males. **c**, AUC-ROC, females ($n = 1,721$). **d**, Sensitivity among the top-risk 10% referred, females. Large markers are the mean across 75 nested cross-validation splits (5 outer folds, 15 seeds); small jittered points are individual splits. Δ AUC and Δ Sensitivity give the gait increment over covariates alone. Conditions are ordered by Δ Sensitivity. All conditions shown reach FDR significance for Δ AUC over covariates alone ($q < 0.1$). Asterisks denote Δ Sensitivity significant at $q < 0.1$ (FDR-corrected). Sensitivity is reported at the top-10% referral threshold, the operating point at which a fixed screening capacity refers the highest-risk decile; pp, percentage points.

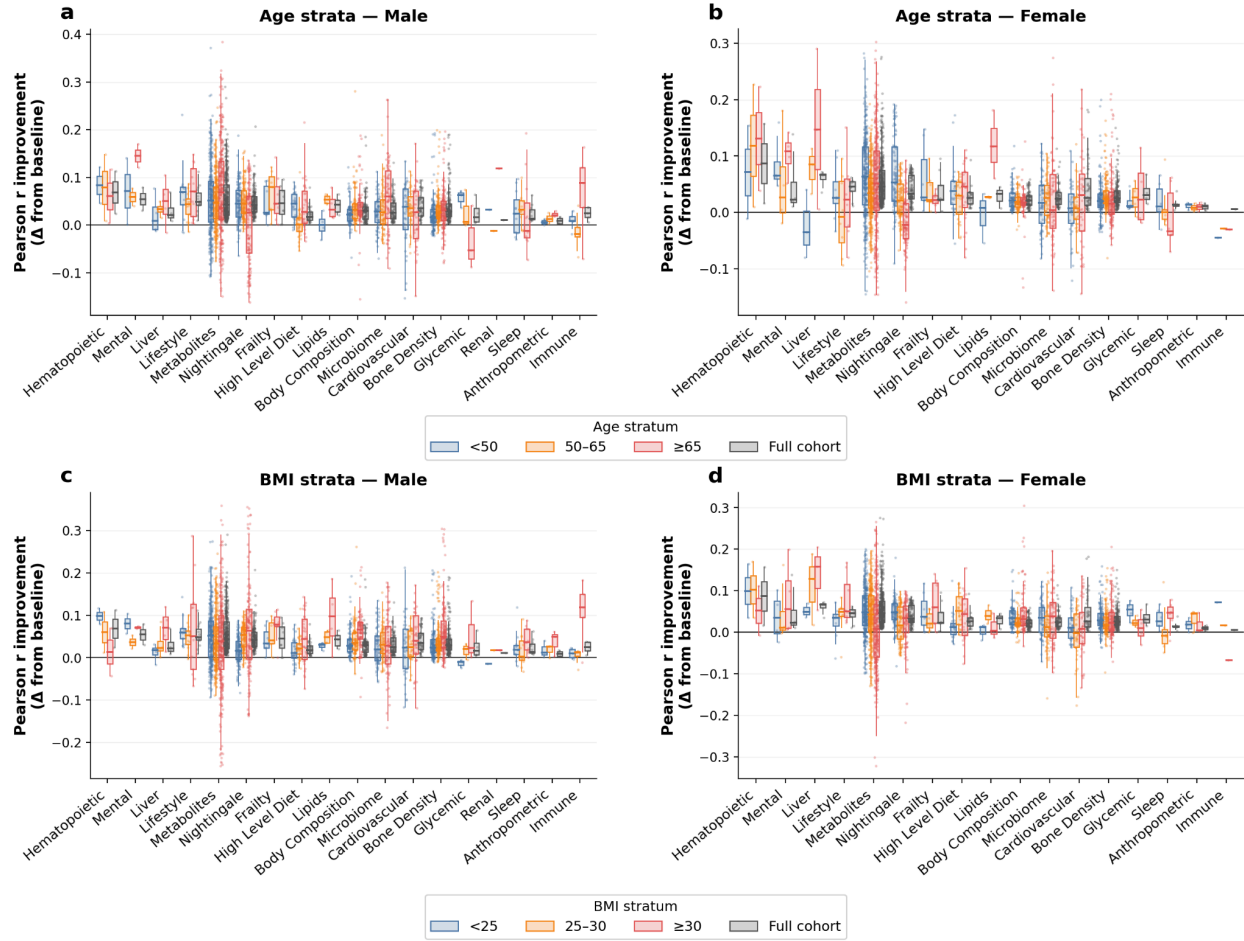


Extended Figure 7: Azure Kinect Body Tracking skeleton model. The 32-joint skeleton output by the Azure Kinect Body Tracking SDK (K4ABT), shown on a reference mannequin. Yellow nodes denote tracked joints; blue arrows indicate the kinematic chain. Six distal hand joints (HAND, HANDTIP, THUMB, bilateral) were excluded due to low detection confidence, yielding a 26-joint skeleton for analysis.

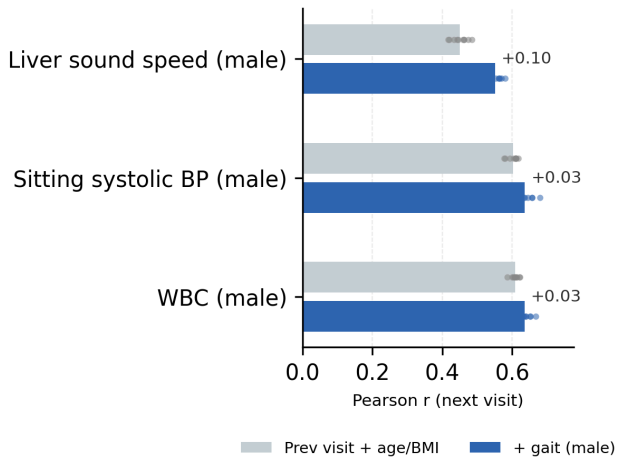
Supplementary



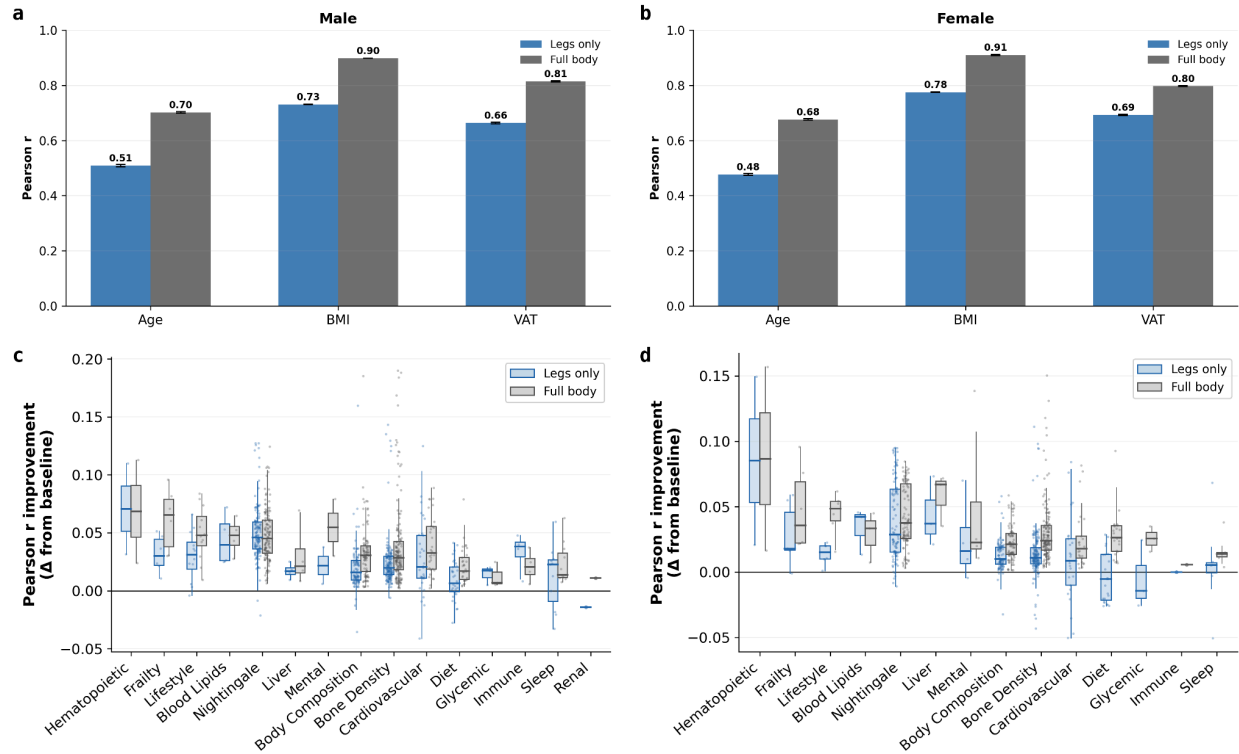
Supplementary Figure 1 | Gait embeddings encode individual identity and are temporally stable. a, Temporal stability of gait embeddings in repeat visitors ($n = 100$ subjects). Within-session similarity (first versus last stride, blue) remains high (mean = 0.86), while across-session similarity (last stride of session 1 versus first stride of session 2, orange) is lower (mean = 0.42) yet well above the random inter-individual baseline (green, centered near zero). Dashed vertical lines indicate distribution means. **b,** Individual specificity of gait embeddings across the full cohort ($n = 3,305$ subject-visits). Distributions of cosine similarity between the first and last stride embeddings of the same individual within a session (orange) versus embeddings from different individuals (blue). Cohen's $d = 3.49$ indicates strong separation, confirming that learned representations capture individual-specific gait signatures. These results demonstrate that individual gait signatures persist across visits despite the expected increase in variability over longer time intervals.



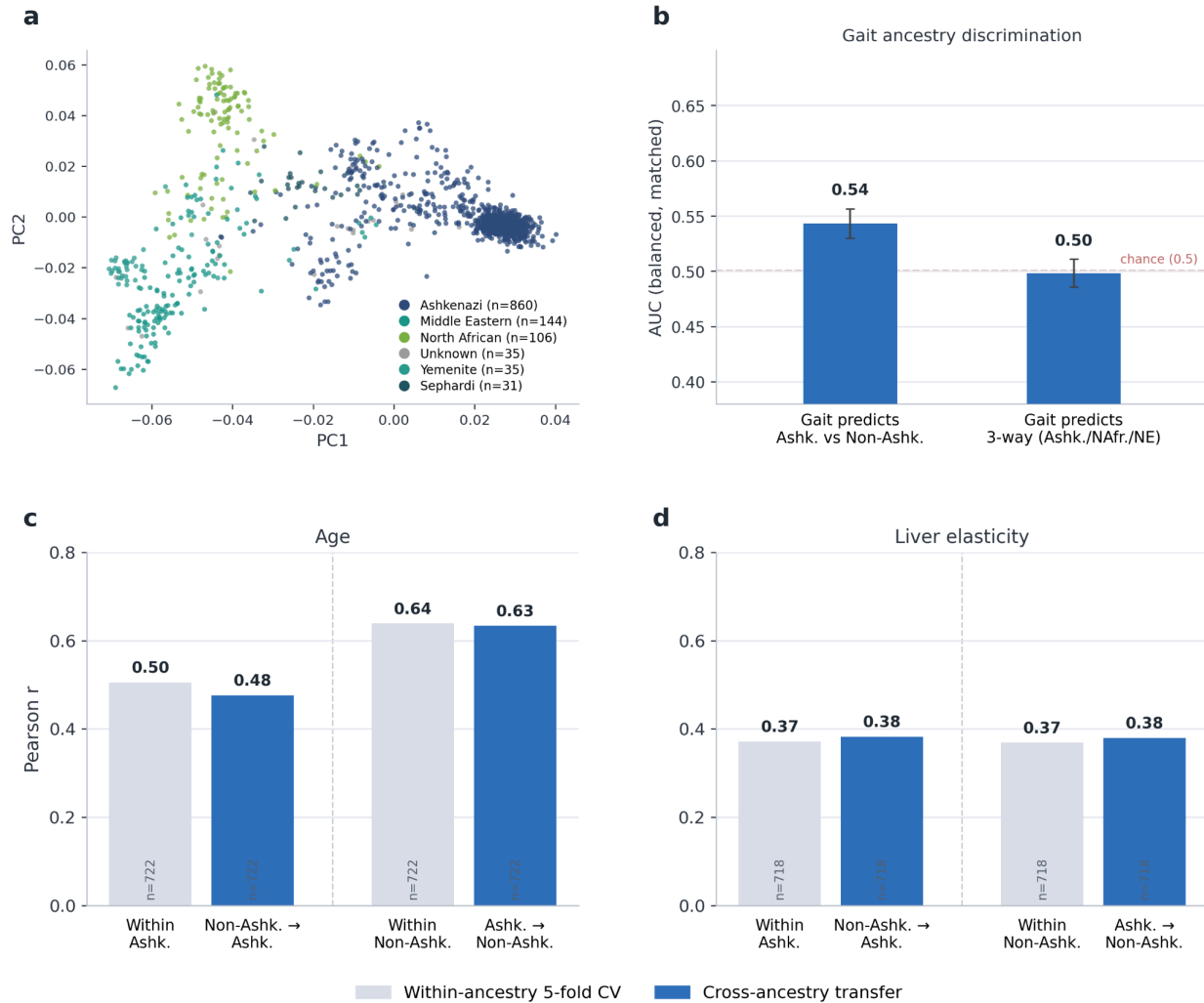
Supplementary Figure 2 | Subgroup stability of gait embedding predictive gains across age and BMI strata. Box and swarm plots showing the predictive gain (Δr) of gait embeddings beyond the demographic and anthropometric baseline (age, BMI, VAT, height), evaluated within clinical subgroups and aggregated at the body-system level (median Δr across FDR-significant features). a,b, Age strata (<50, 50–65, ≥ 65) for male (a) and female (b) participants. c,d, BMI strata (<25, 25–30, ≥ 30) for male (c) and female (d) participants. The full-cohort distribution (grey) is overlaid in each panel for reference. The FDR-significant feature set was frozen from the full-cohort analysis (Figure 4); subgroup evaluation was performed on this fixed feature set without re-selection, isolating subgroup variation in effect size from differences in feature discovery. Body systems are ordered as in Figure 4. Predictive gain was evaluated using 5-fold nested cross-validation repeated across 15 random seeds ($n = 75$ train–test splits), sex-stratified throughout. Box, interquartile range; center line, median; whiskers, $1.5 \times$ IQR; points, individual phenotypes.



Supplementary Figure 3 | Gait embeddings predict biomarkers at the next visit. Horizontal bars show Pearson r between predicted and measured visit-2 phenotype values for a baseline model (age, BMI, VAT, height, and the visit-1 phenotype value; grey) versus the same model with visit-1 gait embeddings added (blue, males). The predictive gain (Δr) attributable to gait is annotated for each phenotype. The analysis was restricted to phenotypes with at least 300 participants having both a visit-1 gait recording and a visit-2 measurement ($n = 554$ participants with repeat visits). Predictions were evaluated using 5-fold nested cross-validation repeated across 15 random seeds ($n = 75$ train-test splits), with sex stratification applied throughout. Labels shown are solely the labels that had significant predicted gains across 15 seeds (FDR $q < 0.1$). Dots show individual cross-validation Pearson r values. Analyses were limited to continuous phenotypes; longitudinal binary outcomes were not modeled due to insufficient incident events.

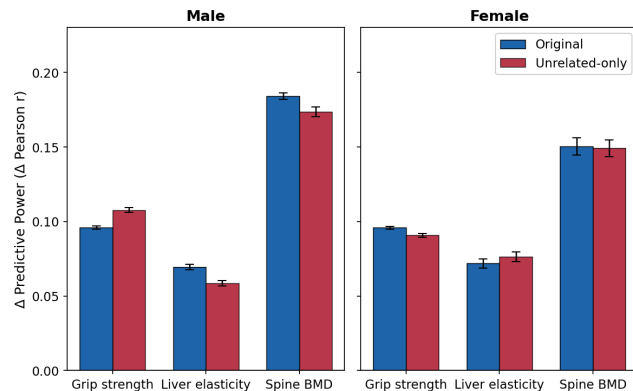


Supplementary Figure 4 | A model pretrained on lower-limb joints alone recovers most of the full-body signal but not its magnitude. a,b, Prediction performance (Pearson r) for age, BMI, and VAT, comparing a lower-limb-only model (blue) against the full-body model (grey), for males (a) and females (b). The lower-limb model was pretrained from scratch on only the 9 lower-limb joints using the same self-supervised architecture, with random masking, and evaluated identically. Bars, mean across 15 seeds; error bars, s.d. Full body led for all three targets, the largest gap being age (male 0.51 vs 0.70; female 0.48 vs 0.68). c,d, Predictive gain (Δr) over the demographic baseline (age, sex, BMI, height, VAT) per body system, for males (c) and females (d), ordered by descending full-body median. Box, IQR; centre line, median; whiskers, $1.5 \times$ IQR; points, individual phenotypes. Lower-limb input reached significance for 91.0% of full-body-significant phenotypes in males and 84.1% in females, but retained less magnitude (per-system median Δr retention 70% male, 48% female).



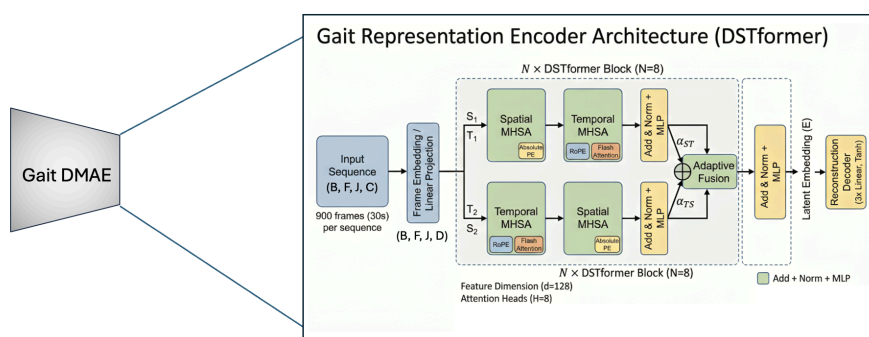
Supplementary Figure 5 | Genetic diversity of the gait cohort, gait embeddings have limited ancestry predictive power, and cross-ancestry transfer of gait-based predictions. a, Genetic principal component analysis of gait cohort participants with available Gencove genotyping data, projected onto the first two principal components (PC1, PC2). Points are colored by genetically inferred ancestry b, Discrimination of genetic ancestry directly from gait embeddings, quantified as the area under the ROC curve (AUC) of classifiers trained to separate ancestry groups on balanced, demographically matched samples (error bars, ± 1 SD across seeds; dashed line, chance = 0.5). Gait embeddings separate Ashkenazi from non-Ashkenazi participants only modestly above chance (AUC = 0.54) and provide essentially no three-way separation among Ashkenazi, North African, and Near Eastern ancestries (AUC = 0.50). c–d, Cross-ancestry transfer of gait-based predictions for age (c; embeddings only) and liver elasticity (d; embeddings combined with age, sex, and BMI). For each target, within-ancestry performance estimated by 5-fold cross-validation (light bars) is compared against cross-ancestry transfer, in which the downstream model was trained on one subpopulation and evaluated on the other (dark bars), in both directions (Non-Ashkenazi to Ashkenazi and Ashkenazi to Non-Ashkenazi). Bars show Pearson r between predicted and measured values; per-bar sample sizes are indicated. Cross-ancestry performance

closely tracks the within-ancestry baseline across targets and directions, indicating that predictive accuracy does not depend on the model having seen the target ancestry during training.



Supplementary Figure 6 | Relatedness within the cohort does not inflate the incremental gait signal.

Gait advantage over baseline (Δ Pearson r , gait+covariates (Age, sender, BMI, VAT) minus covariates-only) for three anchor endpoints. a, Male. b, Female. Original (blue): full cross-validation. Unrelated-only (red): for each fold and seed, test subjects whose relative appeared in that fold's training set were removed before computing r . Bars show mean across 15 seeds; error bars, ± 1 SEM. The gait advantage changed by ≤ 0.011 r in all strata, with no consistent direction, consistent with sampling variation from the reduced test set rather than systematic inflation. 782 related pairs involving 596 participants (17.9% of cohort).



Supplementary Figure 7 | Detailed architecture of the Gait Representation Encoder (DSTformer).

The Dual-Stream Spatiotemporal Transformer (DSTformer, 3.2M parameters) processes input skeleton sequences of shape (B, F, J, C), batch size, 900 frames (30 s at 30 fps), 26 joints, and 4 channels (x, y, z coordinates plus detection confidence), through a frame embedding and linear projection layer, yielding a (B, F, J, D) tensor with feature dimension $D = 128$. The architecture comprises $N = 8$ repeated DSTformer blocks, each containing two parallel attention streams: a spatial-first stream that applies spatial multi-head

self-attention (MHSA) with absolute positional encoding across joints before temporal MHSA with Rotary Position Embeddings (RoPE) and Flash Attention across frames, and a temporal-first stream that reverses this order. Stream outputs are combined via learned adaptive fusion weights (α_{ST} , α_{TS}), followed by residual addition, layer normalization, and a feed-forward MLP. The fused representation passes through a final normalization and MLP layer to produce latent embeddings (E), which serve as input to downstream phenotype prediction models. A reconstruction decoder consisting of three linear layers with Tanh activation maps embeddings back to the original coordinate space during self-supervised pretraining. All attention heads $H = 8$.

Supplementary table 1-3

 [supplementary.xlsx](#)